

FCFS Appointment Simulation: Metric-Focused Sensitivity Report

CUIMC Appointment Simulation

May 9, 2026

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1 Executive Summary

This report is organized by metric. Each metric section asks the same questions:

1. What does the metric mean?
2. Which assumptions move it the most?
3. What is the simplest plot to read?
4. What do the two-class heatmaps add?

Metric	Main drivers	Main warning
Average utilization	No-show risk, then cancellation and demand level	High utilization does not imply good access.
Overall served rate	Total demand, no-show risk, cancellations, balking	This is the main access metric; read it with utilization.
Mean offered wait	Total demand, balking tolerance, cancellations	Offered wait includes patients who later balk.
Balking rate	Balking step and balking threshold	Useful diagnostic; not a success metric by itself.
Class served-rate gap	Differences between class behavior parameters	Class labels matter only when class assumptions differ.

The baseline treats the two classes symmetrically: Class 1 and Class 2 have the same arrival rate, cancellation probability, balking rule, no-show rule, and value. Symmetric heatmaps should therefore be read as absolute parameter effects, not class-label effects. Class advantage appears when the classes receive different behavior parameters.

2 Sensitivity Analysis Methodology

The sensitivity analysis asks how the simulated metrics move when selected input assumptions change around the baseline. It is a model-based stress test, not an empirical causal estimate. All runs use the same FCFS engine and measurement semantics; only the scenario parameters and random seeds vary.

The baseline is `configs/baseline.yaml`: two symmetric classes, 32 slots per day, a 14-day booking horizon, 50 arrivals per class per day, cancellation probability 0.10, balking threshold 9 with high-delay balking probability 0.50, and no-show threshold 6 with high-delay no-show probability 0.30. The low balking and no-show probabilities are 0.00. For behavior rules, step means high minus low, so the baseline balking step is 0.50 and the baseline no-show step is 0.30.

Each sensitivity block answers four questions: what changes, what it changes around, what stays fixed, and what hypothesis the plot is meant to test.

2.1 Arrival Load and Mix

Class-specific arrival slices change one class's `lambda_per_day` around the baseline value of 50 arrivals per day. The slice range is 25 to 75 arrivals per day, or 0.5 to 1.5 times baseline, while the other class and all behavior rules stay fixed. Arrival-mix heatmaps change total daily arrivals from 50 to 150, or 0.5 to 1.5 times the baseline total of 100, and change Class 1's share from 0.1 to 0.9. FCFS stress

curves use a wider total arrival range of 30 to 170, or 0.3 to 1.7 times baseline, at the baseline 50/50 mix. These runs test the demand-pressure hypothesis: as arrivals rise against fixed capacity, served rate should fall and offered wait should rise, while utilization may stay high.

2.2 Balking Step

The balking-step grids change the high-delay rejection probability increase, high minus low. The baseline is 0.50, and the grid ranges from 0.00 to 1.00 in 21 evenly spaced values. In one-driver slices, Class 1 changes while Class 2 stays at 0.50. In two-class heatmaps, Class 1 and Class 2 move independently over the same range. These runs test whether stronger high-delay rejection lowers access for the affected class and whether lower accepted wait can be a selection effect caused by long-delay patients leaving.

2.3 Balking Threshold

The balking-threshold grids change the offered delay at which high balking begins. The baseline is 9 days, and the grid uses all possible thresholds in the 14-day horizon, $0, 1, \dots, 13$. Threshold 13 means the high balking probability never applies to offered delays inside the horizon. These runs test whether greater waiting tolerance increases the affected class's served rate, and whether that added tolerance also keeps more patients in the system long enough to raise offered wait.

2.4 No-Show Step

The no-show-step grids change the high-delay no-show probability increase. The baseline is 0.30, and the grid ranges from 0.00 to 1.00 in 21 evenly spaced values. One-driver slices vary Class 1 while Class 2 stays at baseline; heatmaps vary both classes. These runs test whether higher delay-related no-show risk reduces completed visits, lowers utilization, and moves the served-rate gap against the class with higher no-show risk.

2.5 No-Show Threshold

The no-show-threshold grids change the accepted delay at which high no-show risk begins. The baseline is 6 days, and the range is $0, 1, \dots, 13$. These runs test whether postponing high no-show risk improves completed-service outcomes and whether earlier high no-show risk primarily shows up as lower utilization, lower access, or a class gap.

2.6 Cancellation Probability

The cancellation grids change each class's start-of-day future-appointment cancellation probability around the baseline 0.10. The range is 0.00 to 0.30 in 21 evenly spaced values. These runs test whether after-booking losses reduce completed service for the affected class and whether cancellations can shorten offered wait by freeing future capacity while still worsening access.

2.7 Common Threshold-Jump Grids

The threshold-jump figures vary a shared threshold and shared jump level for both classes together. The jump level ranges from 0.00 to 1.00, while thresholds range from 0 to 13. These are not class advantage tests; they test how the shape of a symmetric threshold rule changes aggregate utilization, access, and offered wait.

2.8 Balking Deep Dives

The Class 1 high-balking probability sweep changes only Class 1’s high balking probability over 0.0, 0.1, $\dots$, 0.9, centered against the baseline value 0.50. The Class 1 balking-threshold sweep changes only Class 1’s threshold over 3, 4, $\dots$, 13, centered against the baseline threshold 9. Class 2 and all non-balking assumptions stay fixed. Each point uses 100 seeds. These runs test the selection hypothesis directly: accepted wait can decrease because long-delay Class 1 patients reject appointments, even when Class 1 access gets worse.

2.9 Regression Screen

The randomized regression screen samples 240 parameter settings with random seed 9137. For each setting, the simulation is run with seeds 4101 and 4102, and the two results are averaged. Sampled inputs are total arrivals from 0.4 to 1.7 times baseline, Class 1 arrival share from 0.1 to 0.9, class-specific balking steps, no-show steps, balking thresholds, no-show thresholds, and cancellation probabilities. Features are converted into average levels and Class 1 minus Class 2 gaps. This screen tests whether the main visual conclusions still appear when many assumptions vary at once. Significance is handled with `statsmodels` OLS using HC3 robust standard errors; the regression figure and summary table keep coefficients with robust p-values below 0.05.

The generated report figures come from `scripts/generate_metric_analysis_figures.py`. The main two-class grids use a fixed seed, 2027, for each grid point so changes across a plot are attributable to parameter changes rather than changing random streams. The scenario comparison averages seeds 2027, 2028, and 2029. The FCFS capacity-stress curves also average those three seeds.

3 Reading Guide and Baseline

3.1 Baseline and Scenario Summary

Baseline uses `configs/baseline.yaml`: $\lambda_1 = 50$, $\lambda_2 = 50$, 32 slots per day, a 14-day booking horizon, balking threshold 9, balking high probability 0.50, no-show threshold 6, no-show high probability 0.30, and cancellation probability 0.10 for both classes.

Scenario	Utilization	Overall served	Accepted wait	Offered wait	Class gap	Delay gap
Baseline	0.839	0.269	8.35	9.30	0.001	0.003
Scenario 2	1.000	0.395	4.29	5.06	-0.008	0.005

Table 1: Scenario values are rates or averages. Scenario 2 changes several assumptions at once, so it is a comparison point, not a one-parameter causal test.

Class gap is `access_advantage_class_1`. Positive values mean Class 1 has the higher served rate. Delay gap is `delay_advantage_class_1`. Positive values mean Class 1 has the lower offered wait because the metric is Class 2 offered wait minus Class 1 offered wait.

3.2 Figure Reading Guide

The plot order is intentional. Driver plots are simple one-axis checks: Class 1 changes on the x-axis while Class 2 stays fixed at its baseline value. Each driver plot shows the overall metric plus the Class 1 and Class 2 values when that class-level value is defined. Heatmaps are the more detailed view: the x-axis changes Class 1, the y-axis changes Class 2, and the dashed baseline line marks the current assumption when applicable.

Color family	Meaning
Blue	Arrival pressure, arrival mix, and demand-load changes.
Purple	Balking step, balking threshold, and balking-rate diagnostics.
Green	No-show step and no-show threshold changes.
Red	Cancellation probability changes.
Line style and marker	Overall, Class 1, and Class 2 within the same driver family.
Gray dashed line	Baseline assumption in one-axis slices.
Black/white dashed crosshair	Baseline Class 1 and Class 2 assumptions in two-axis heatmaps.
Driver-colored heatmaps	The varied assumption family. Overall metric panels use a sequential version of the driver color. Class-gap panels use a muted driver-adjacent diverging scale centered on a near-white zero; one side is Class 2 advantage and the other side is Class 1 advantage, as labeled by the colorbar sign.

Table 2: Color key used throughout the metric-focused figures.

Behavior step and jump level both refer to the probability increase $\text{high}_i - \text{low}_i$ after the threshold. Threshold axes are in days of offered delay.

3.3 Measurement Semantics

This report preserves the current simulation semantics. These conventions are important for interpreting the metric values and are not model changes.

- Class-level patient metrics are tracked for patients who arrive during the measurement window.
- Slot utilization is tracked for service days during the measurement window.
- If `cooldown_days < horizon_days - 1`, some late measurement-window bookings can remain unresolved after the simulation ends. These patients are currently not counted as served, canceled, or no-show.
- `total_value` is service-day based. It increments when a patient is served, including untracked burn-in or cooldown bookings, so it is not cohort-aligned with tracked class metrics.
- The engine can support more than two classes, but this report and its class-gap metrics are explicitly two-class analyses.

4 Metric Results

4.1 Average Utilization

Definition

Average utilization measures completed visits per available slot:

$$\text{average_utilization} = \frac{1}{D} \sum_d \frac{\text{served slots on day } d}{\text{slots per day}}.$$

No-shows do not count as utilization because the slot did not become a completed visit. Overall utilization is aggregate. In the driver plot, the Class 1 and Class 2 lines decompose utilization into each class's share of all available slots.

Sensitivity Readout

No-show behavior is the clearest direct driver: booked slots disappear without becoming visits. Cancellation can also change utilization because it changes the appointment pool before service. Demand pressure is more subtle. Utilization can stay high across a wide demand range, even when the share of patients served collapses.

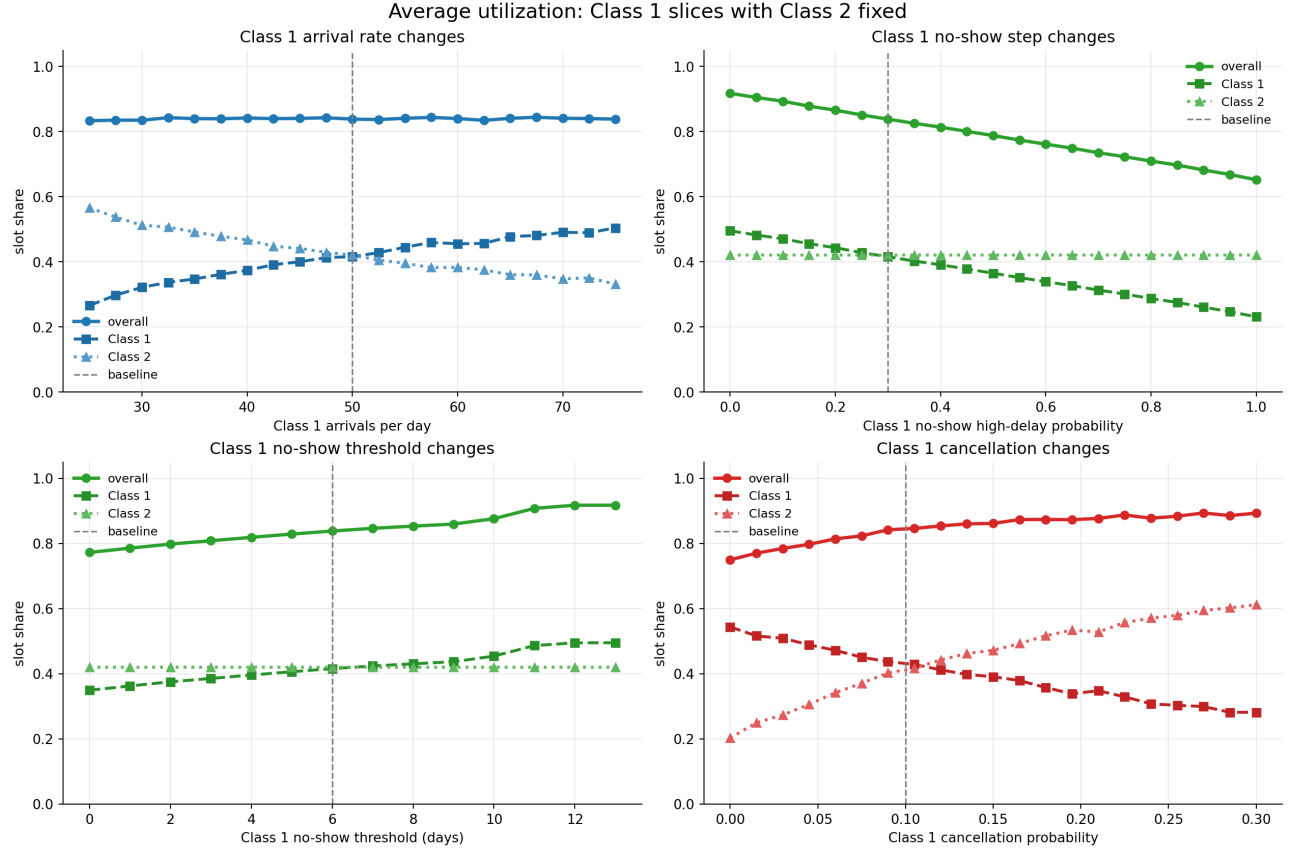


Figure 1: Average utilization driver plots. Class 1 varies while Class 2 stays fixed at baseline. The class lines are each class's served slots divided by all available slots.

The simple view shows that utilization is not the same as access. Around baseline demand, the clinic is busy, but only a small share of arrivals are served. No-show assumptions move utilization more directly than demand once the system is already capacity constrained.

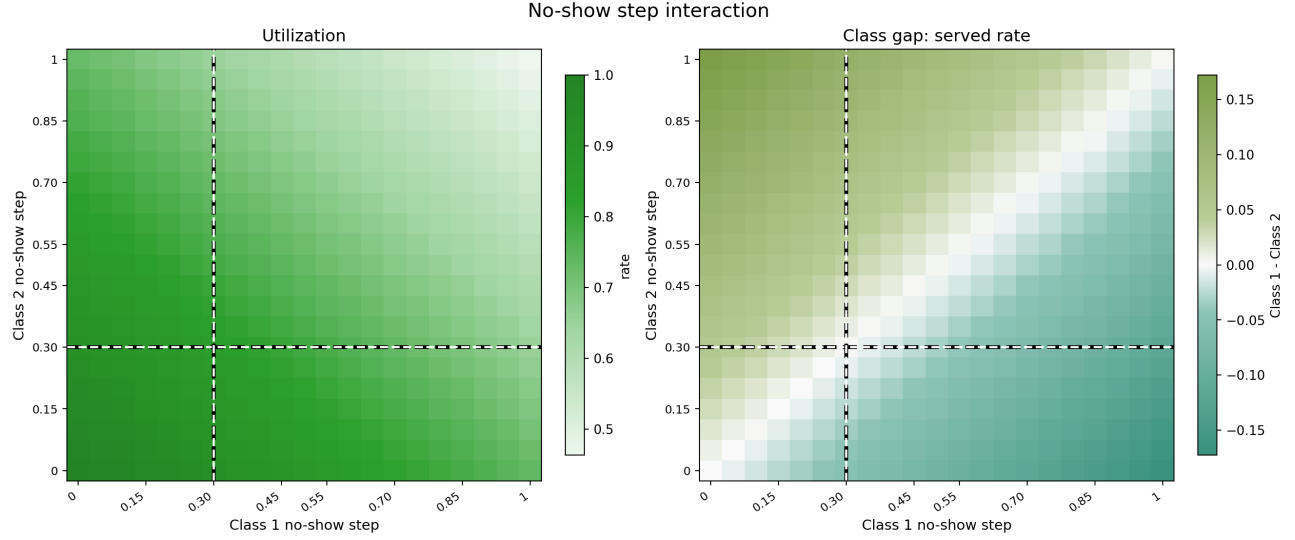


Figure 2: No-show step heatmap. Left: utilization. Right: Class 1 served-rate gap.

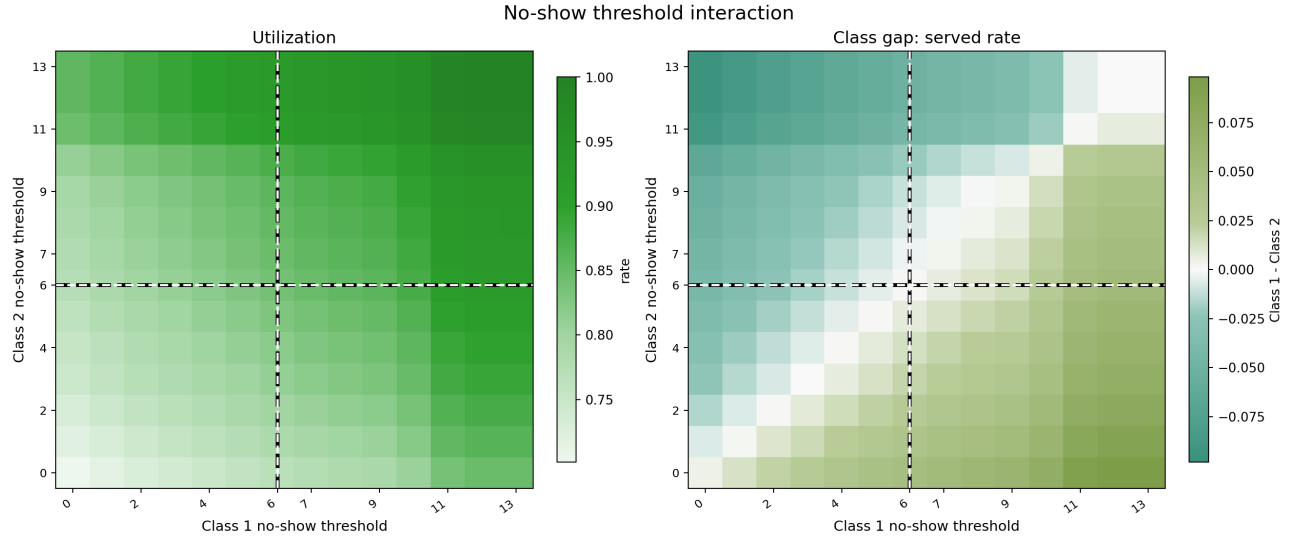


Figure 3: No-show threshold heatmap. Left: utilization. Right: Class 1 served-rate gap.

The heatmaps add the class-specific reading. When both classes have high no-show risk, utilization falls. When only one class has higher no-show risk, that class loses completed visits and the served-rate gap moves against it.

4.2 Overall Served Rate

Definition

Overall served rate is the main access metric:

$$\text{overall_percent_serviced} = \frac{\sum_i V_i}{\sum_i A_i},$$

where A_i is arrivals and V_i is served patients for class i .

Sensitivity Readout

The strongest aggregate driver is total arrival pressure. More arrivals compete for the same FCFS booking horizon and same daily capacity. No-shows and cancellations lower completed visits after booking. Balking lowers served rate because patients who reject long-delay offers leave the system.

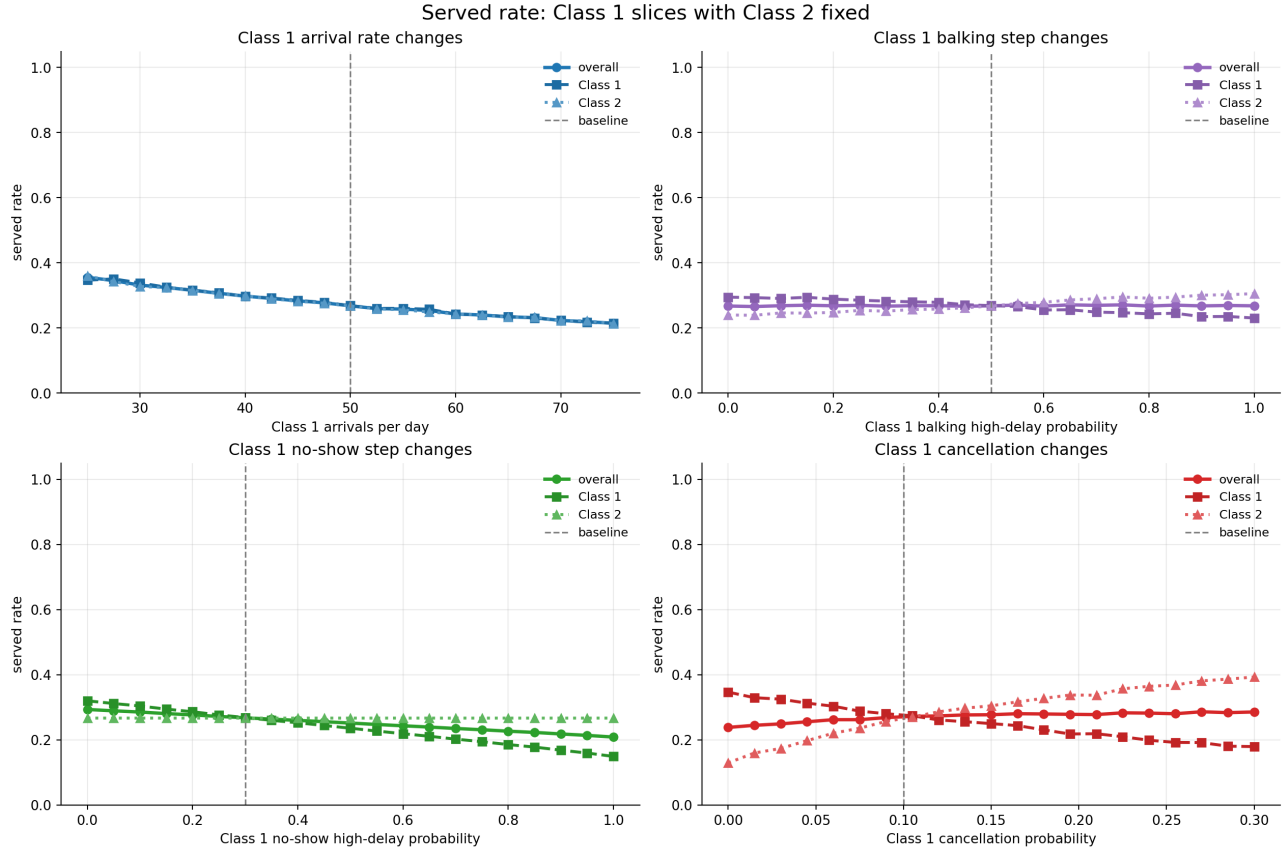


Figure 4: Served-rate driver plots. Class 1 varies while Class 2 stays fixed at baseline; each panel shows overall, Class 1, and Class 2 served rates.

The easiest reading is that served rate falls when demand exceeds capacity. At low demand nearly every arrival is served; near baseline the clinic remains highly utilized but access is already poor.

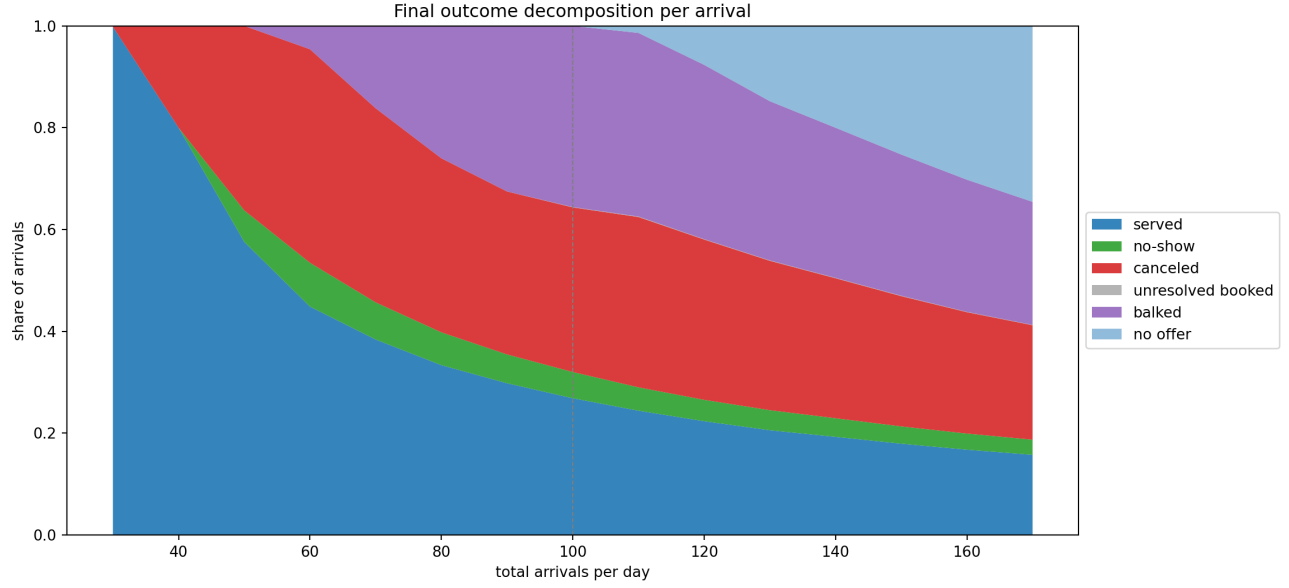


Figure 5: Final outcome decomposition under demand pressure. Each band is a share of arrivals.

The outcome decomposition explains why served rate falls. At baseline, many arrivals are lost to balking or after-booking losses. Under heavier overload, **no_offer** becomes a major failure mode because the FCFS horizon fills.

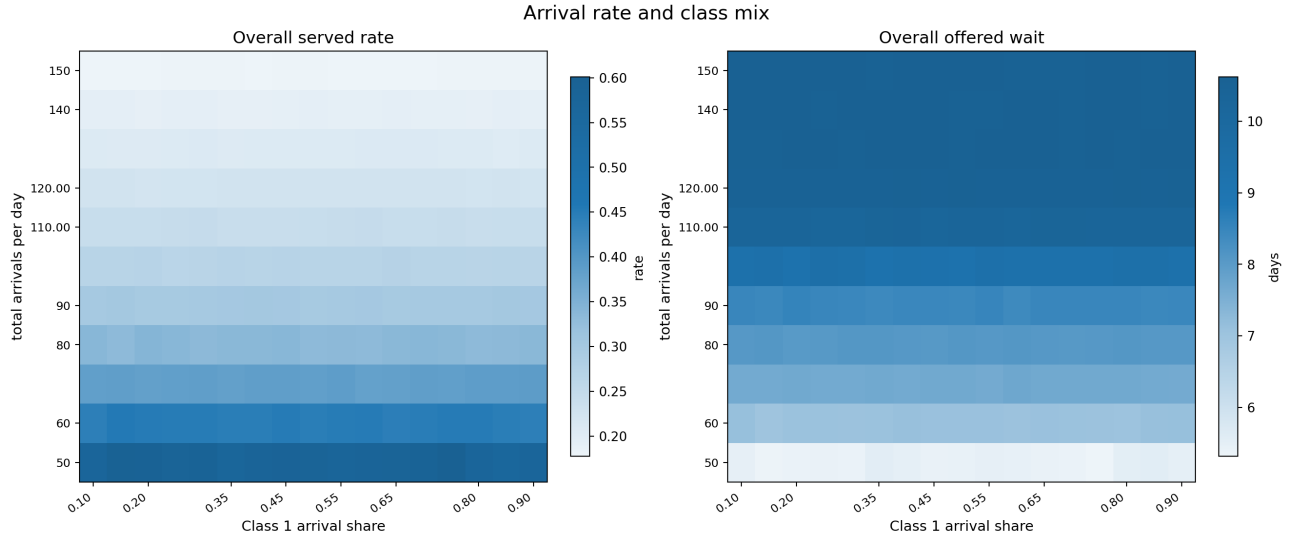


Figure 6: Arrival-rate and class-mix heatmap. Left: overall served rate. Right: offered wait.

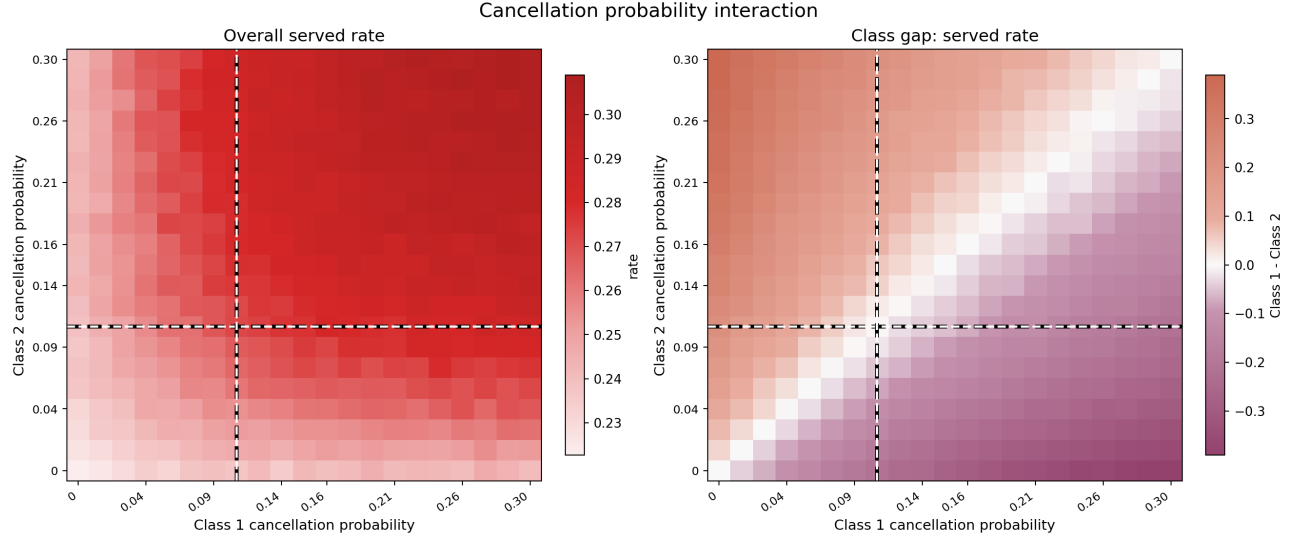


Figure 7: Cancellation heatmap. Left: overall served rate. Right: Class 1 served-rate gap.

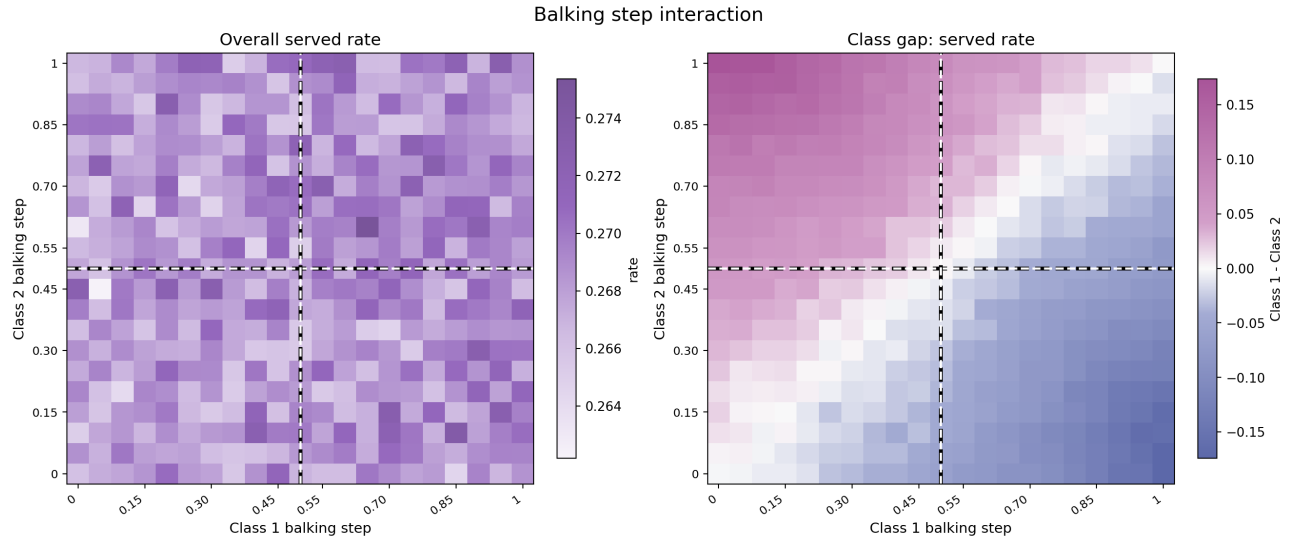


Figure 8: Balking step heatmap. Left: overall served rate. Right: Class 1 served-rate gap.

The heatmaps show two different mechanisms. Arrival pressure mostly changes the aggregate served rate. Cancellation and balking differences can change both aggregate access and which class receives more completed visits.

4.3 Mean Offered Booking Delay

Definition

Mean offered booking delay is the patient-facing wait metric:

$$\text{mean_offered_booking_delay} = \frac{\sum_i \text{total offered delay}_i}{\sum_i O_i},$$

where $O_i = B_i + K_i$ is offered patients: accepted bookings plus balked patients. Patients with `no_offer` are excluded because they never receive a slot offer.

Sensitivity Readout

Demand pressure raises offered wait because patients are pushed farther into the booking horizon. Balking behavior changes wait by changing whether long-delay offers remain in the accepted appointment pool. Cancellations can shorten offered wait by freeing future capacity, but they do so by losing already-booked patients.

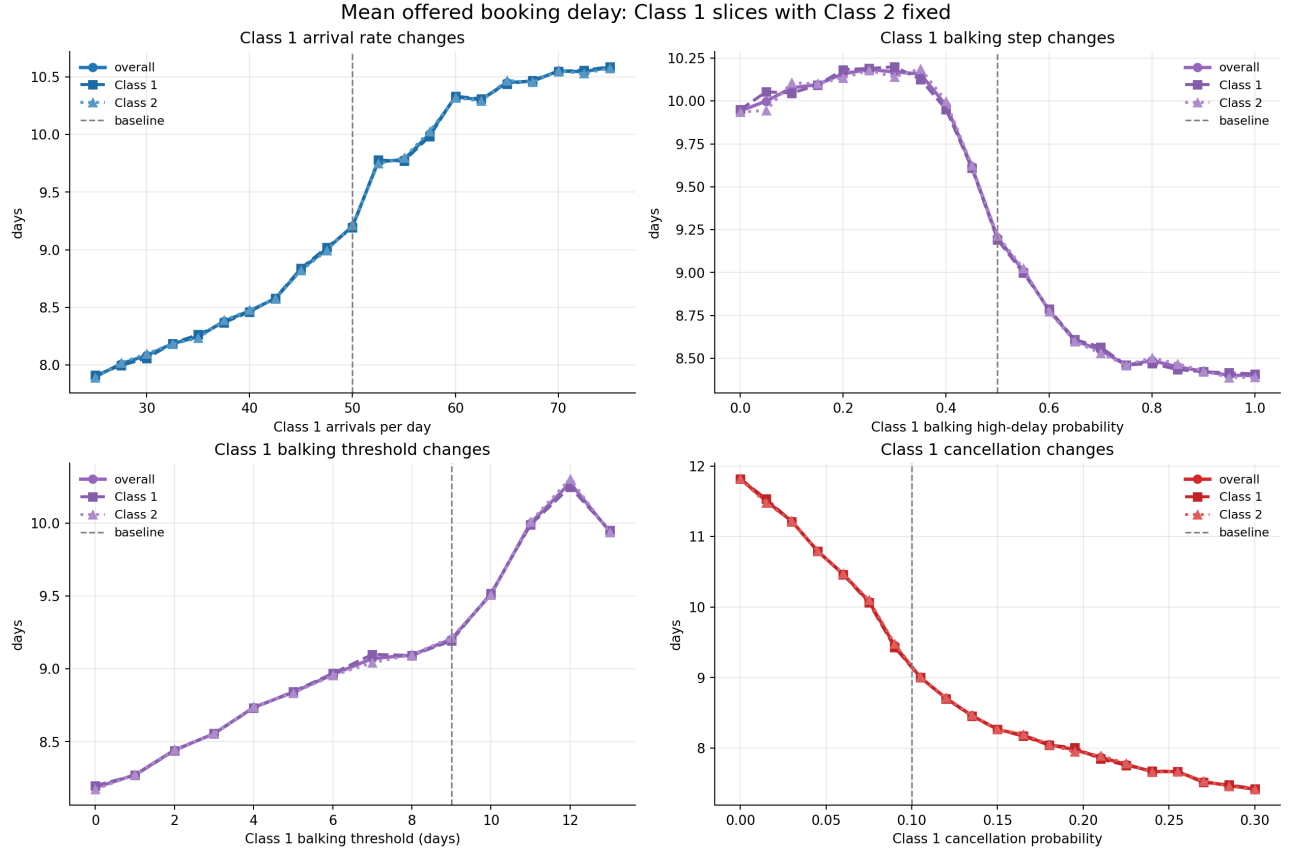


Figure 9: Mean offered-wait driver plots. Class 1 varies while Class 2 stays fixed at baseline; each panel shows overall, Class 1, and Class 2 offered waits.

The simple view shows that offered wait is most intuitive under demand pressure: more arrivals push offers farther out. Balking and cancellation need more care because shorter waits can be caused by patients leaving the system, not by better access.

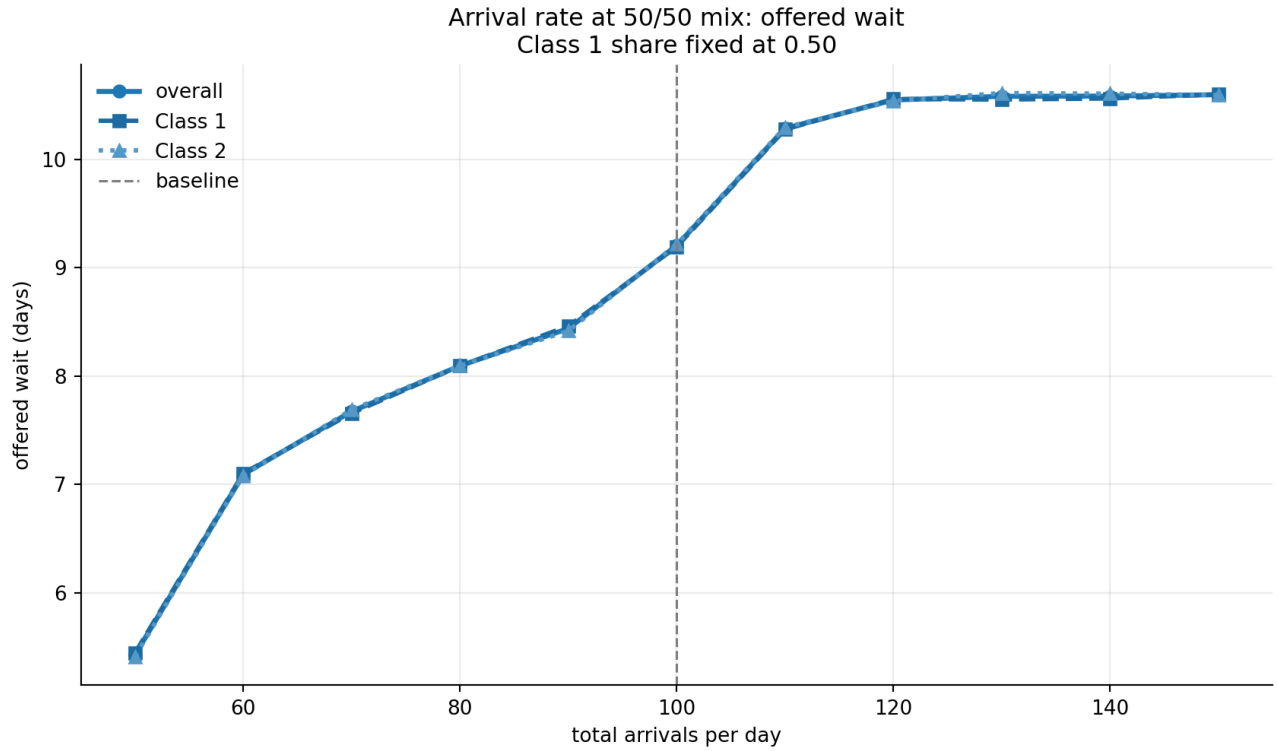


Figure 10: Offered-wait slice as total arrivals change and class mix stays 50/50.

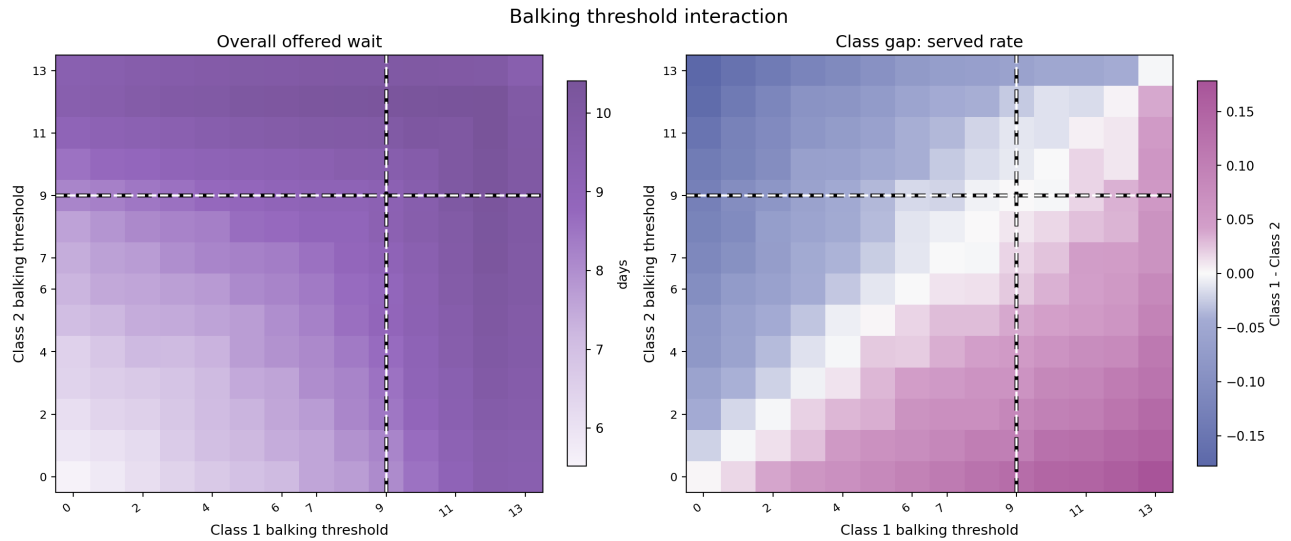


Figure 11: Balking threshold heatmap. Left: offered wait. Right: Class 1 served-rate gap.

The balking-threshold heatmap adds the class interpretation. The class with the higher tolerance threshold accepts longer waits and is more likely to stay in the system. That can increase its served rate while also keeping the system more congested.

4.4 Balking Rate

Definition

Balking rate is a diagnostic for rejected offers:

$$\text{balking_rate} = \frac{\sum_i K_i}{\sum_i O_i}.$$

It is measured among patients who received an offer. No-offer patients are excluded because they never had the opportunity to accept or reject a delay.

Sensitivity Readout

Balking rate is controlled directly by the balking step and threshold. A higher step makes high-delay offers more likely to be rejected. A lower threshold makes that high rejection probability start earlier.

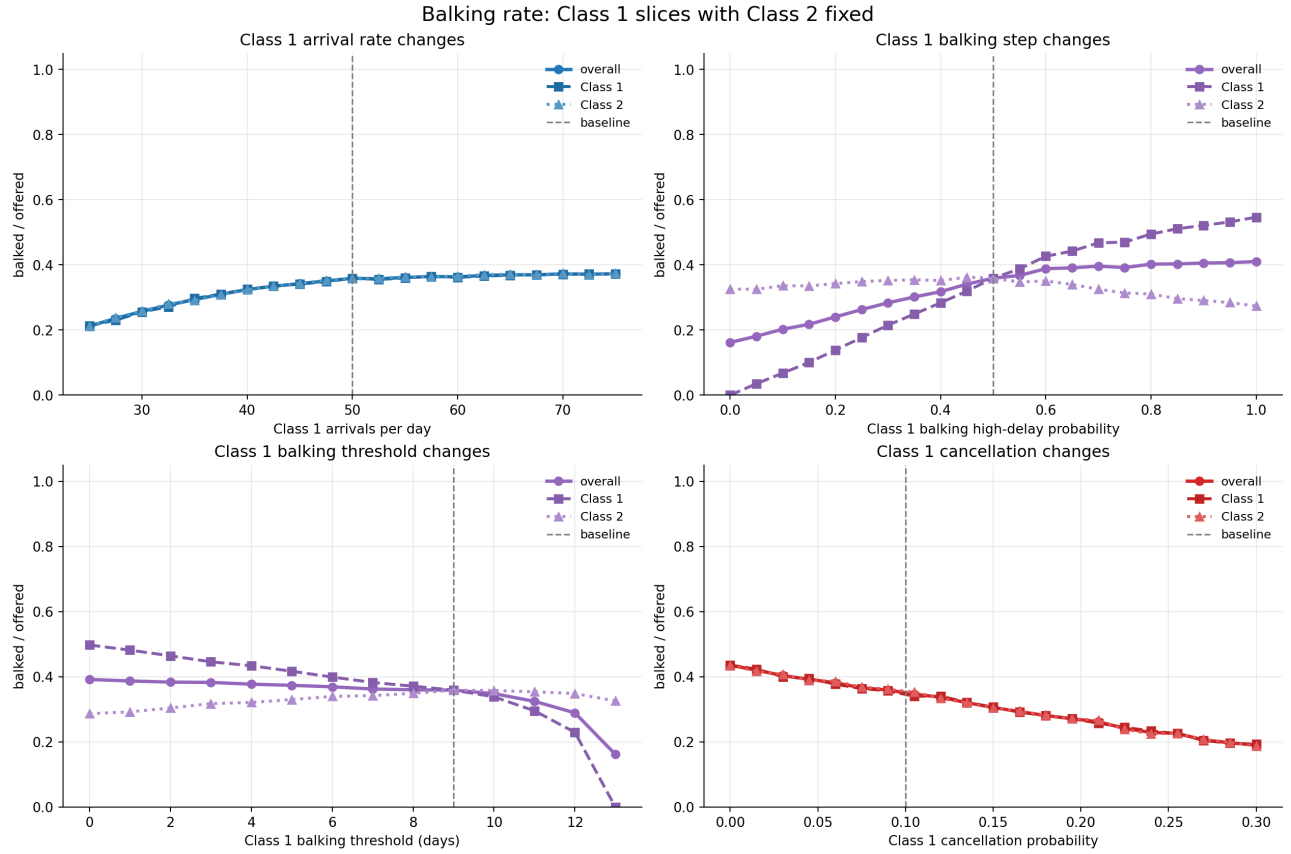


Figure 12: Balking-rate driver plots. Class 1 varies while Class 2 stays fixed at baseline; each panel shows overall, Class 1, and Class 2 balking rates.

Balking rate helps explain why a lower wait is not always an improvement. If wait falls because many long-delay offers are rejected, access can worsen.

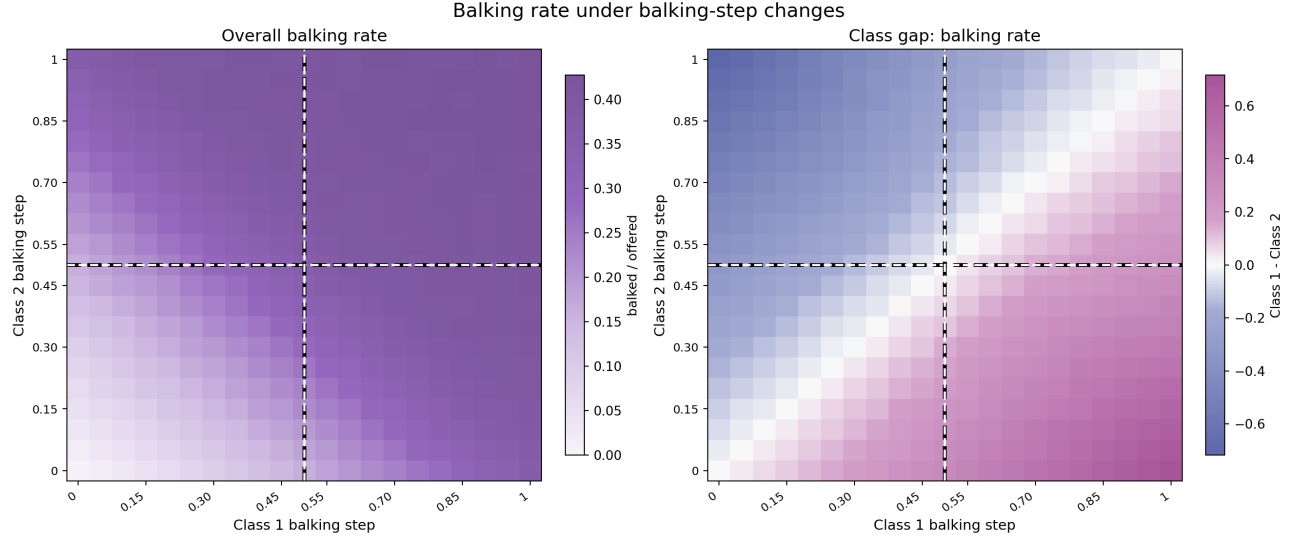


Figure 13: Balking-rate heatmap under balking-step changes. Left: overall balking rate. Right: Class 1 minus Class 2 balking-rate gap.

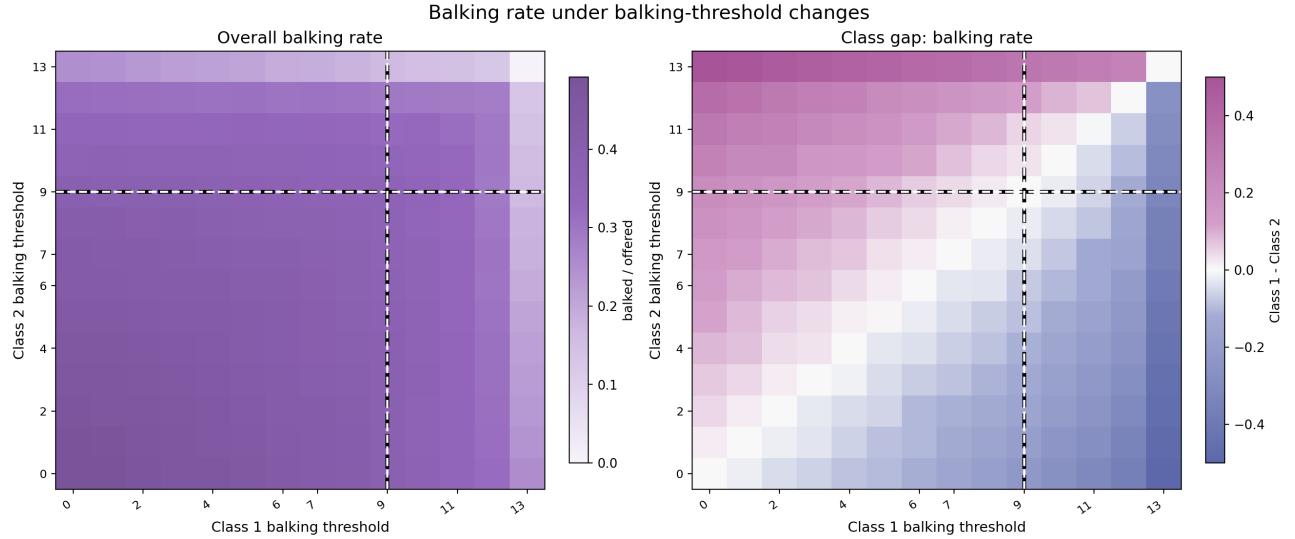


Figure 14: Balking-rate heatmap under balking-threshold changes. Left: overall balking rate. Right: Class 1 minus Class 2 balking-rate gap.

The heatmaps separate absolute balking from class differences. When both classes have high balking, overall balking rises. When one class has higher balking than the other, the more patient class tends to receive more service.

4.5 Class Served-Rate Gap

Definition

Class served-rate gap is the main class-benefit metric:

$$\Delta_{\text{access}} = \text{percent_served}_1 - \text{percent_served}_2.$$

Positive values mean Class 1 is served more often. Negative values mean Class 2 is served more often.

Sensitivity Readout

Class gap is driven by differences between class assumptions. In the current FCFS baseline, class labels are otherwise exchangeable. Cancellation gaps are the largest observed class-access driver, followed by balking threshold gaps, balking step gaps, and no-show gaps.

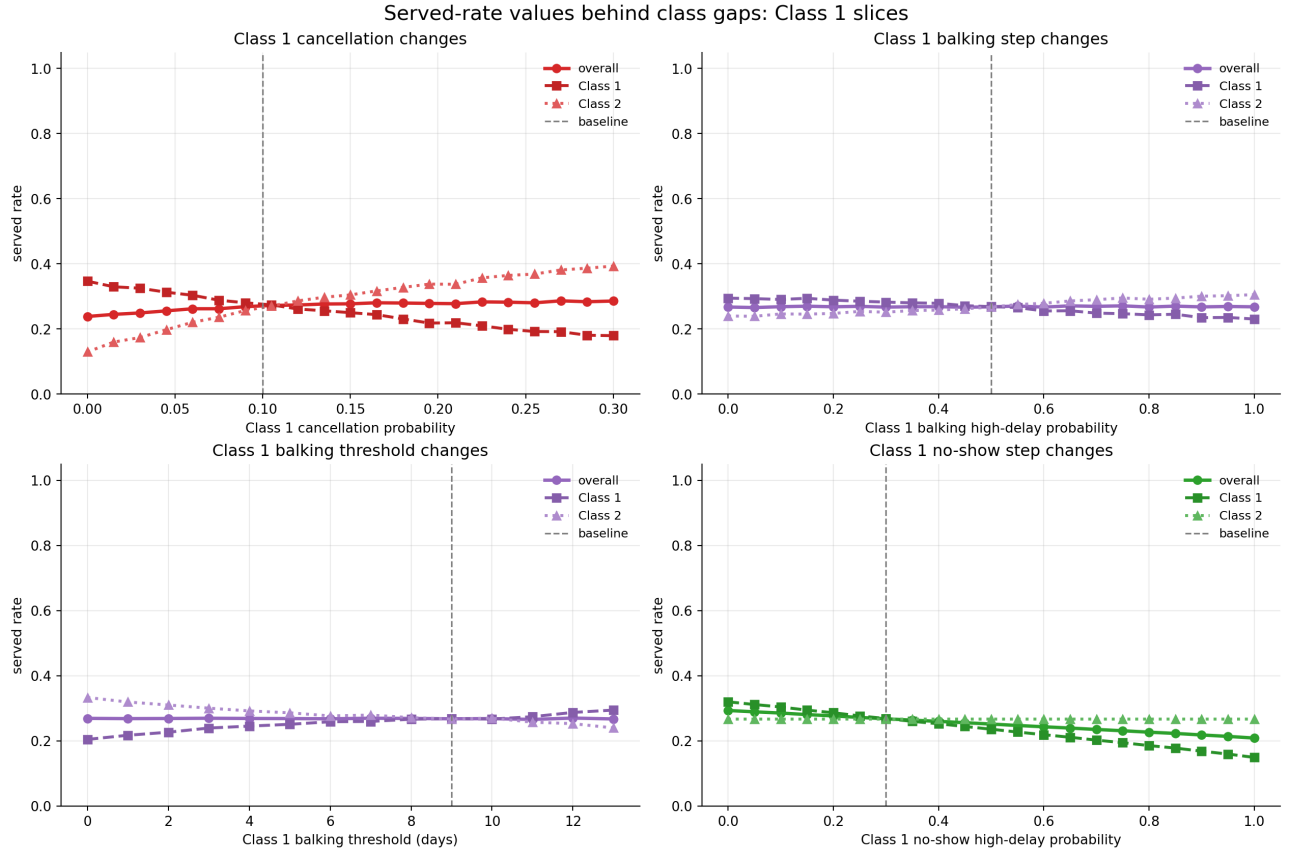


Figure 15: Served-rate values behind class gaps. Class 1 changes while Class 2 stays fixed at baseline. The class gap is the vertical distance between the Class 1 and Class 2 lines.

The simple class-gap plots show direction through the distance between class lines. Higher Class 1 cancellation probability, higher Class 1 balking step, and higher Class 1 no-show step move the Class 1 line below the Class 2 line. A higher Class 1 balking threshold helps Class 1 because it tolerates longer offered waits.

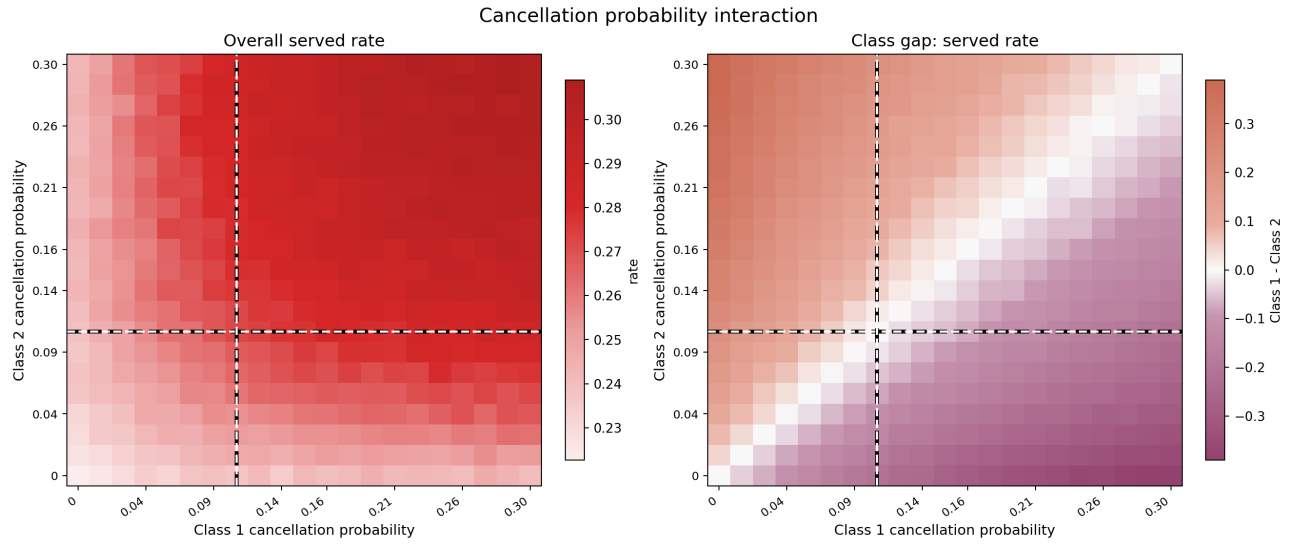


Figure 16: Cancellation interaction heatmap. The right panel shows the served-rate gap created by cancellation differences.

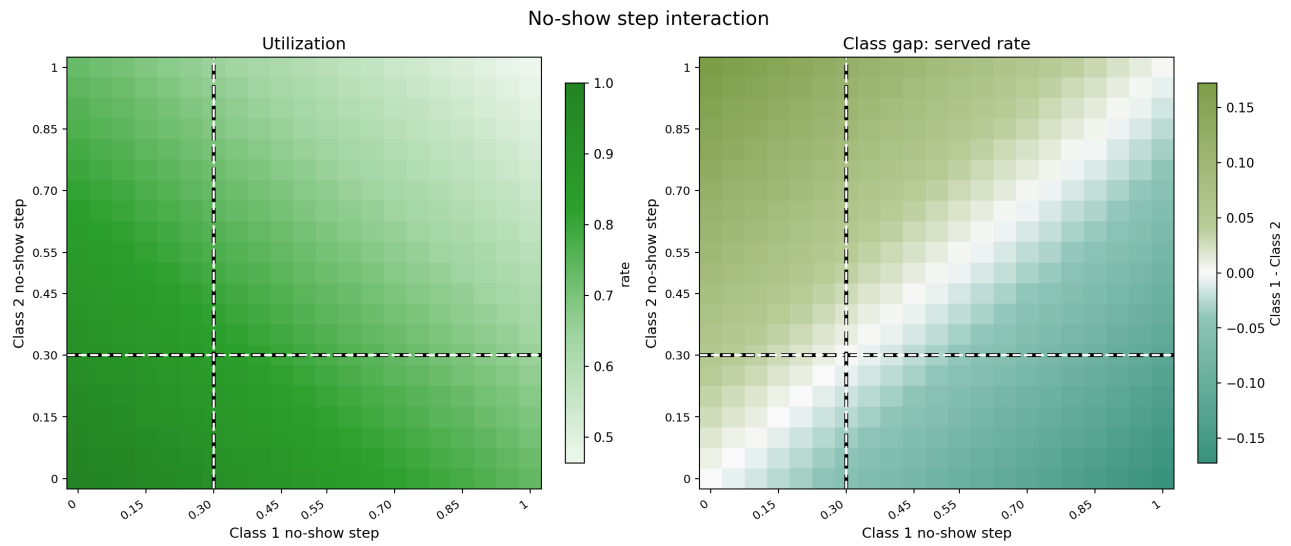


Figure 17: No-show step interaction heatmap. The right panel shows the served-rate gap created by no-show differences.

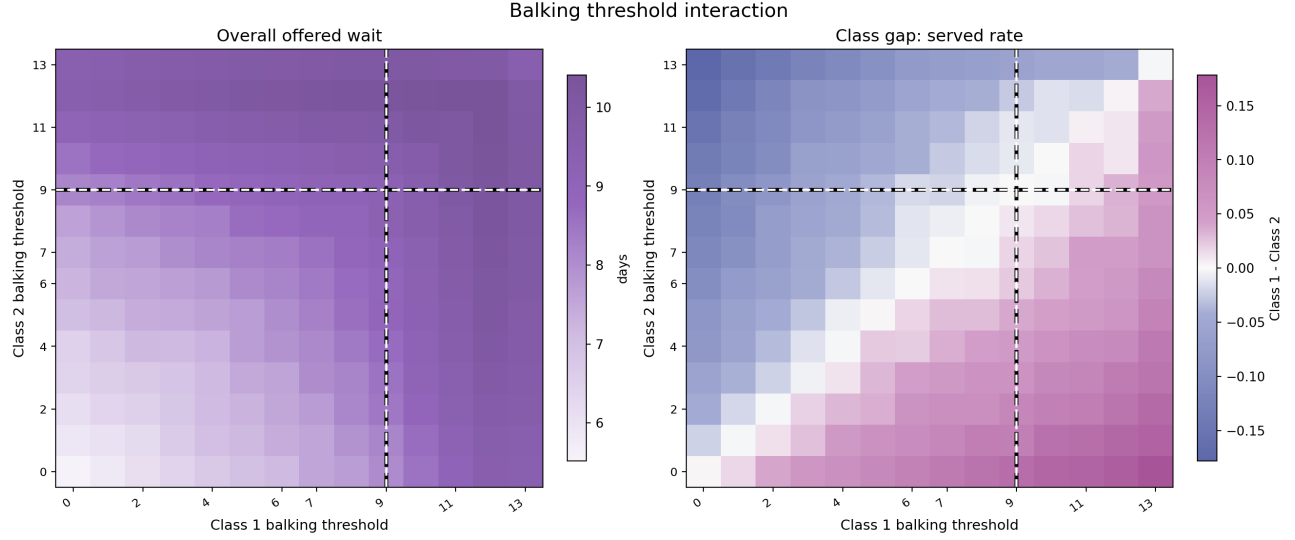


Figure 18: Balking threshold interaction heatmap. The right panel shows the served-rate gap created by tolerance differences.

The heatmaps show symmetry around the baseline because the baseline classes are identical. Moving to the upper-left or lower-right corners creates opposite class advantages.

5 Balking Deep Dive

The balking deep dive isolates Class 1 balking behavior while Class 2 stays at baseline. This section is intentionally narrower than the main metric sections: it explains why a lower accepted delay can be a selection effect rather than a patient-access improvement.

5.1 Class 1 High Balking Probability

This sweep varies only the high balking probability for Class 1. All other simulation parameters, including Class 2 behavior, stay fixed.

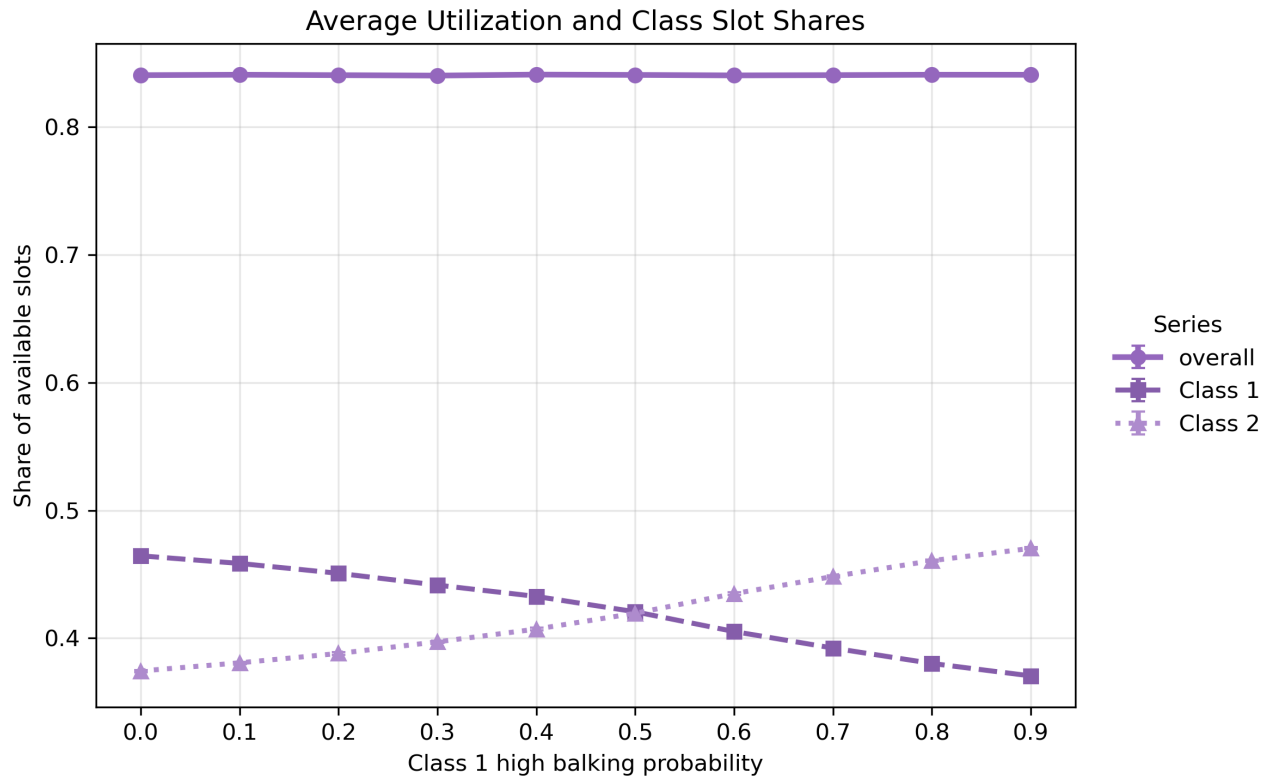


Figure 19: Class 1 high-balking-probability sweep: aggregate utilization.

Aggregate utilization remains nearly flat across Class 1 high balking probabilities. Total demand remains high enough to keep capacity used even when more Class 1 patients reject long-delay offers.

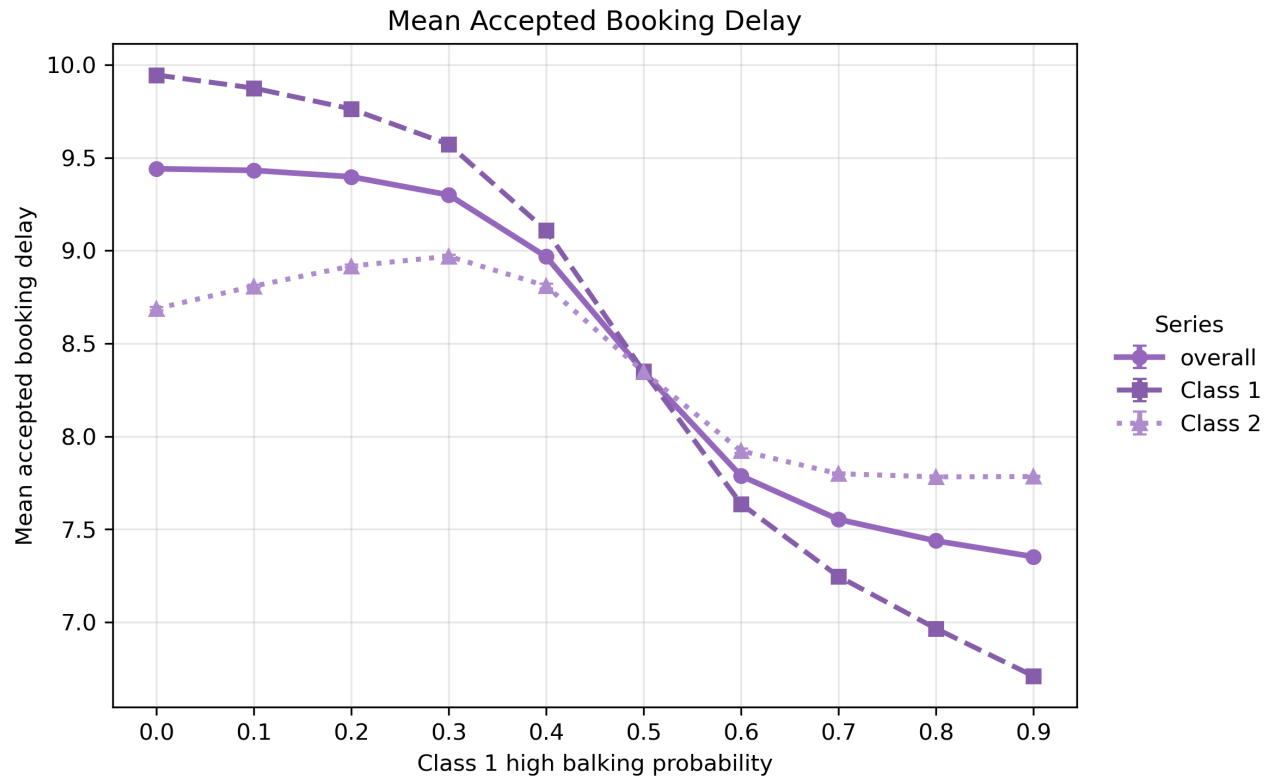


Figure 20: Class 1 high-balking-probability sweep: mean accepted delay by class.

Class 1 accepted delay declines as high balking probability increases because long-delay Class 1 patients increasingly reject appointments and leave the accepted sample.

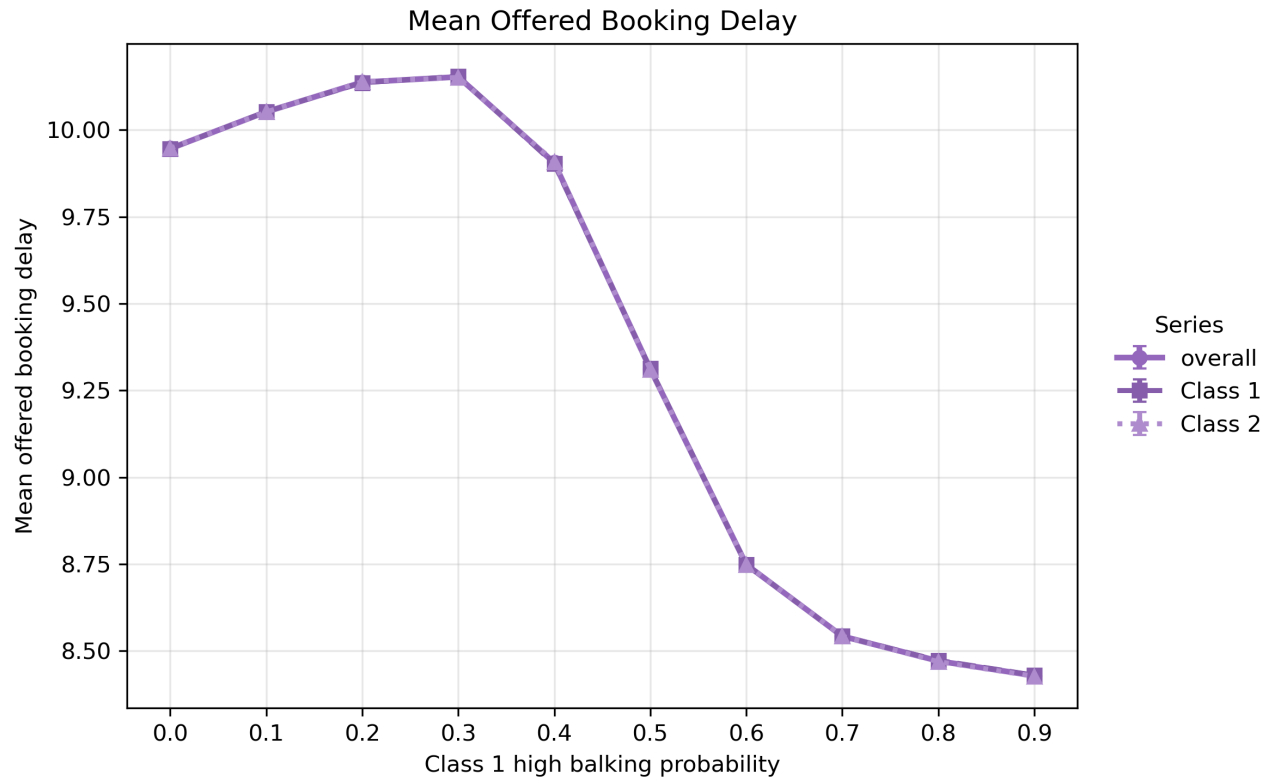


Figure 21: Class 1 high-balking-probability sweep: mean offered delay by class.

Offered delay is the broader congestion measure because it includes both accepted and balking patients. Once enough long-delay Class 1 patients reject offers, congestion falls and both classes can receive shorter offered delays.

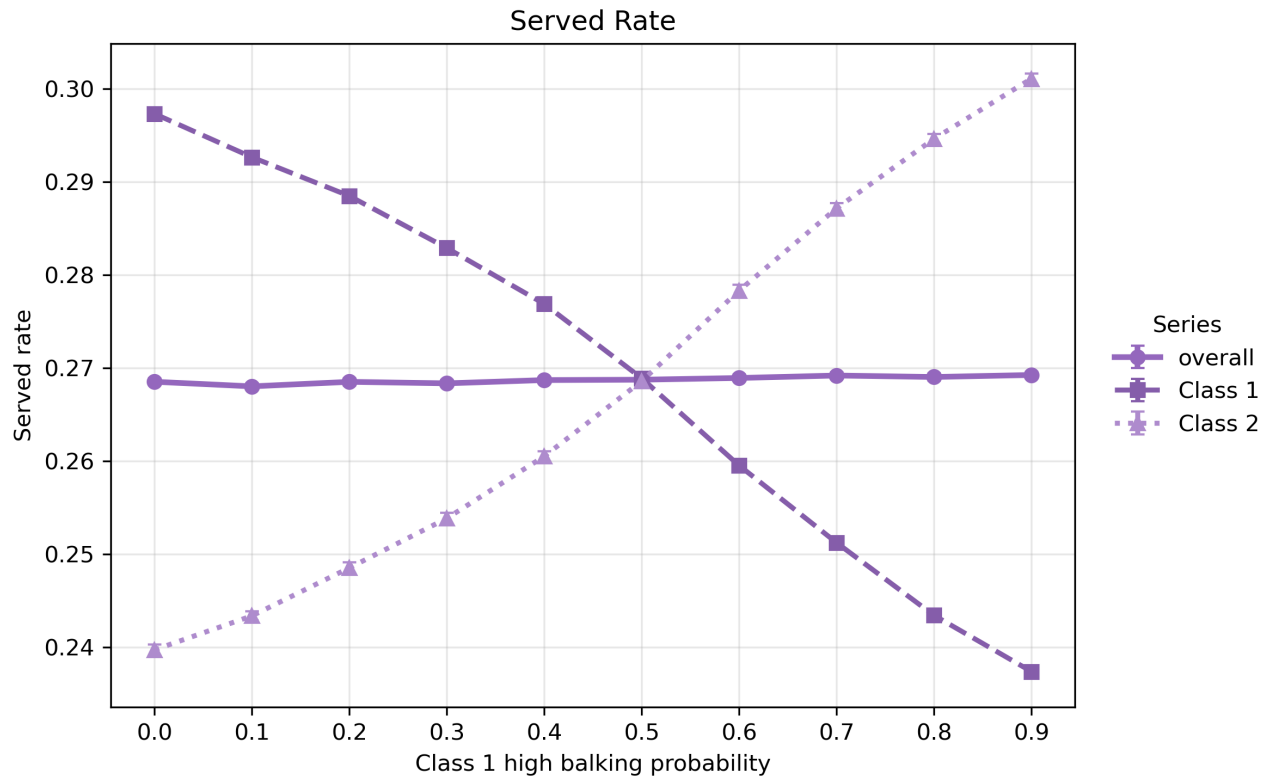


Figure 22: Class 1 high-balking-probability sweep: served rate by class.

Percent serviced shows the class tradeoff: as Class 1 high balking probability rises, Class 1 access falls while Class 2 access rises.

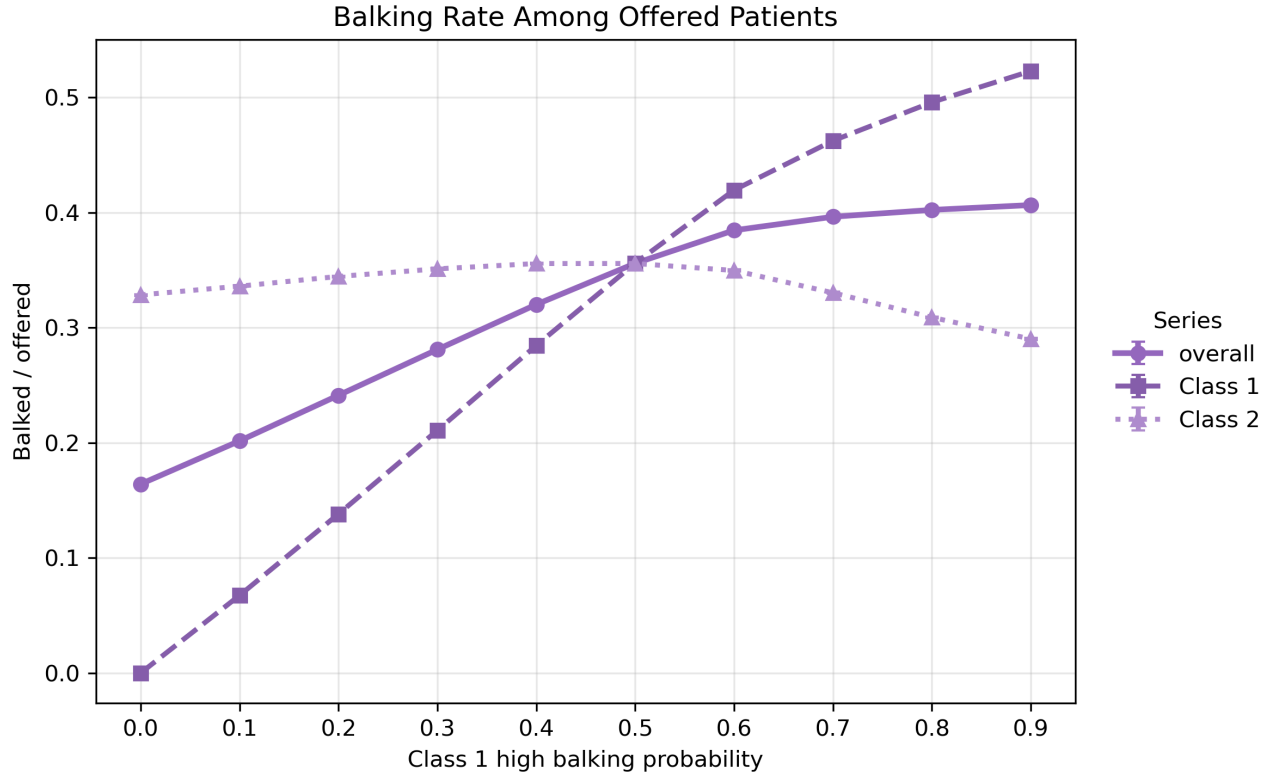


Figure 23: Class 1 high-balking-probability sweep: balking rate by class.

The balking-rate plot confirms that this is a behavioral rejection mechanism, not just a capacity effect.

5.2 Class 1 Balking Threshold

This sweep varies only the Class 1 balking threshold. Class 1's low and high balking probabilities remain fixed. With `horizon_days = 14`, possible offered delays are $\tau = 0, 1, \dots, 13$.



Figure 24: Class 1 balking-threshold sweep: aggregate utilization.

Aggregate utilization should be read as completed slot use. If the curve is flat, the threshold mainly redistributes access rather than changing total completed capacity.

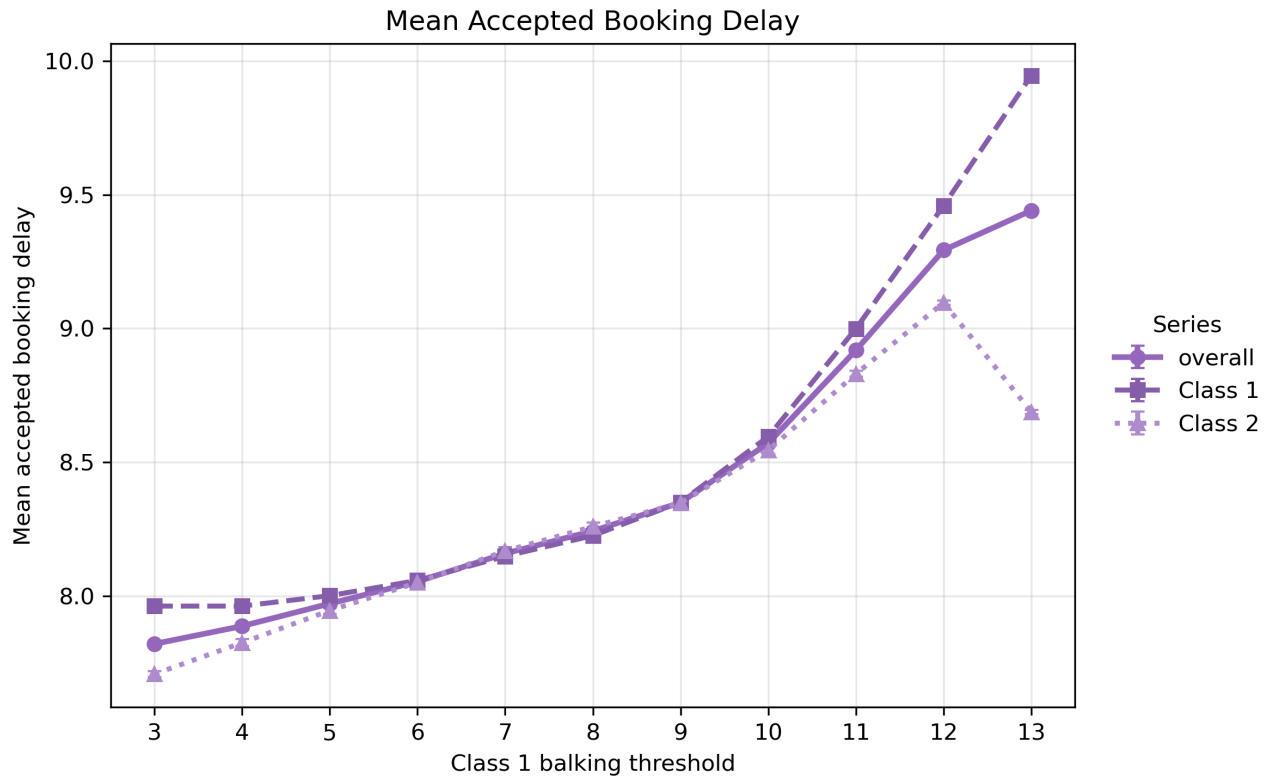


Figure 25: Class 1 balking-threshold sweep: mean accepted delay by class.

Class 1 accepted delay generally rises as the threshold increases because Class 1 tolerates longer waits before high balking applies.

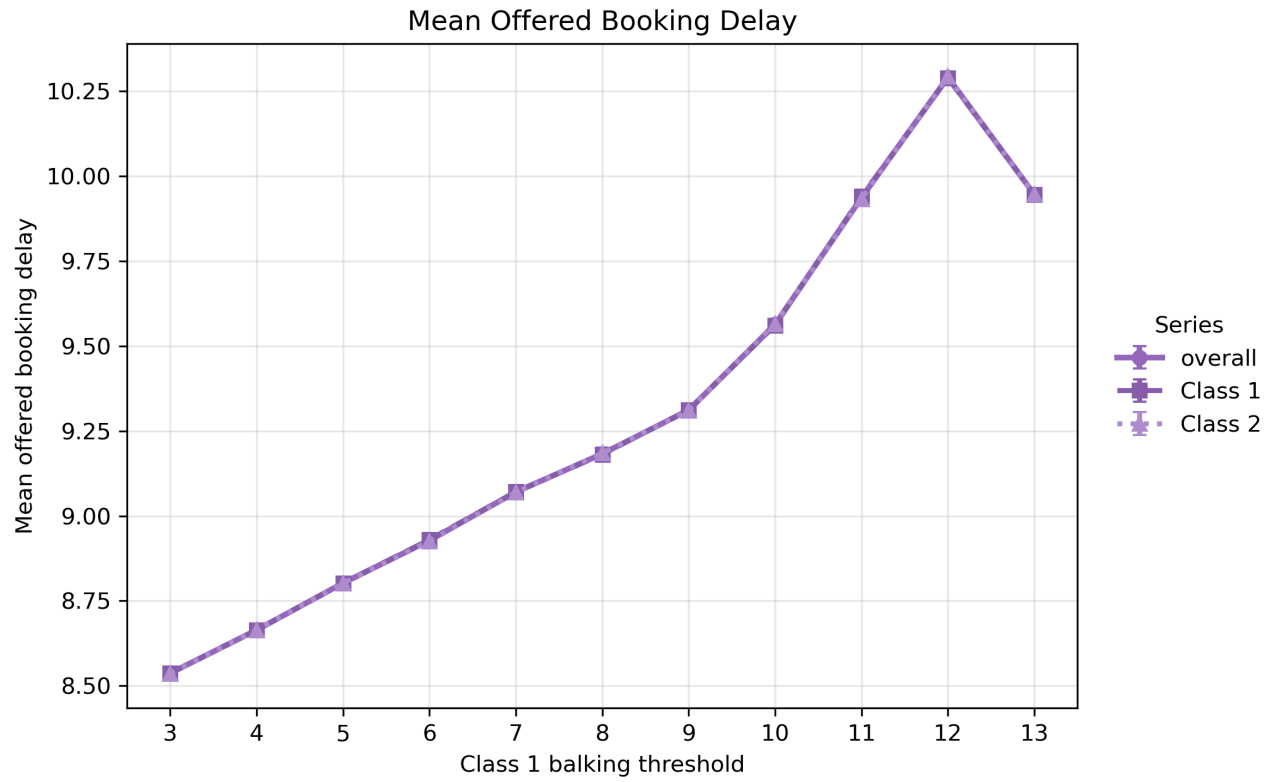


Figure 26: Class 1 balking-threshold sweep: mean offered delay by class.

Mean offered delay shows how the booking horizon changes when Class 1 begins balking earlier or later.

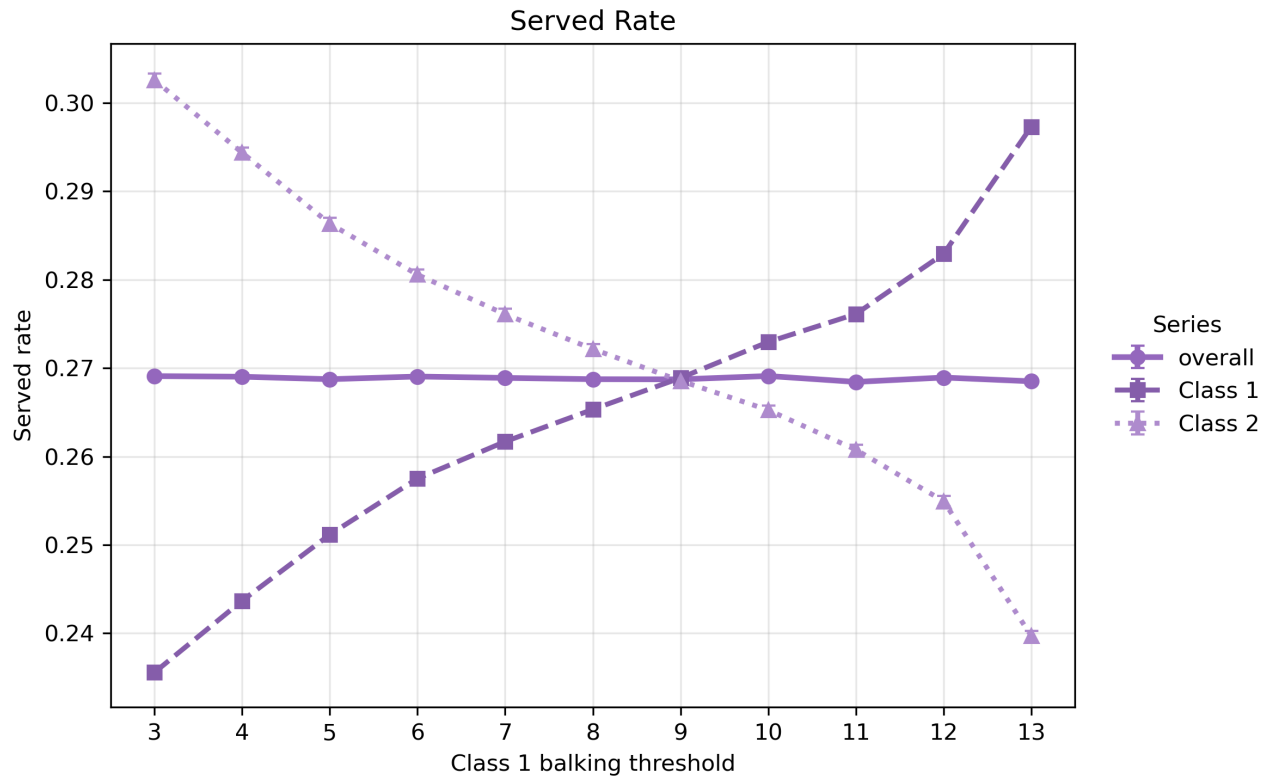


Figure 27: Class 1 balking-threshold sweep: served rate by class.

Percent serviced is the access metric that determines whether shorter accepted waits are actually good for patients.

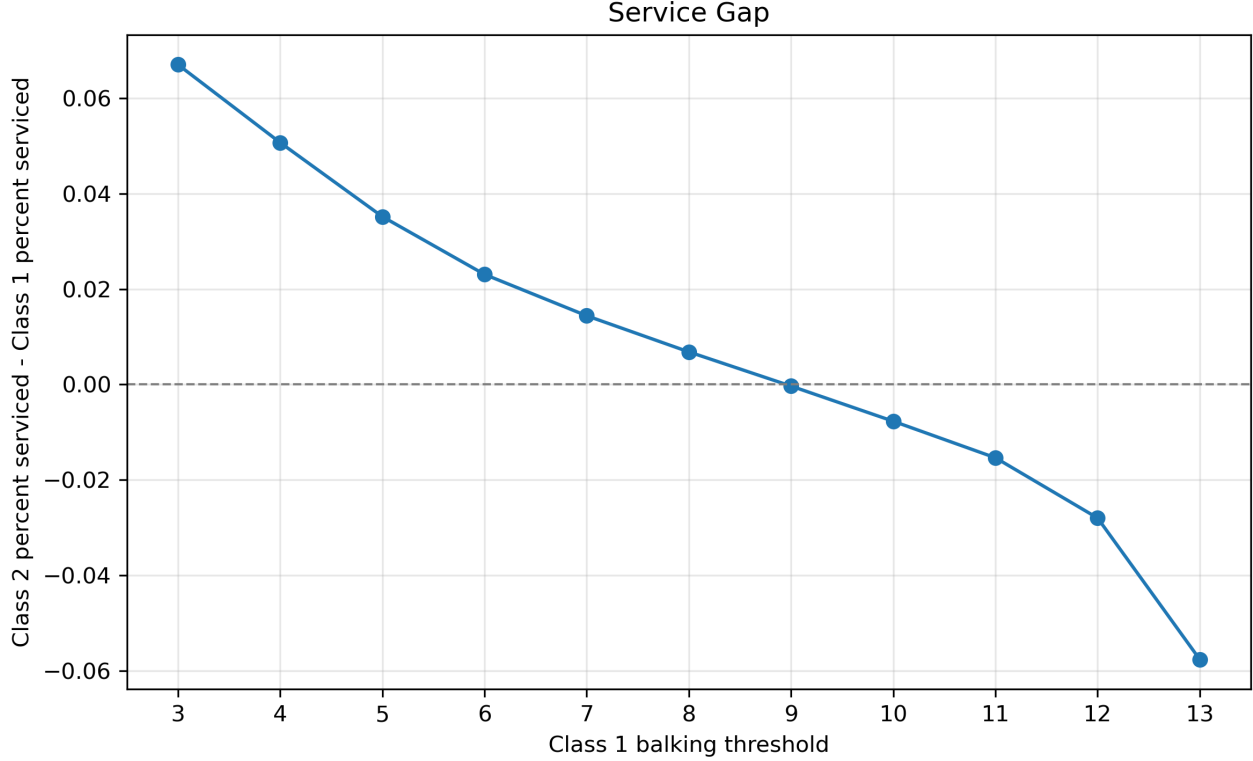


Figure 28: Class 1 balking-threshold sweep: Class 1 minus Class 2 service gap.

The service-gap figure directly shows whether changing the threshold primarily redistributes service between classes.

6 Regression Screen

The regression screen handles significance explicitly. All 12 standardized coefficients for each target are saved in `outputs/reports/metric_analysis/data/regression_standardized_coefficients.csv` with HC3 robust standard errors, 95% robust confidence intervals, robust p-values, and a `significant_05_hc3` flag. The figure and table below keep only coefficients with HC3 robust p-values below 0.05.

For each of 240 randomized FCFS parameter settings, the simulation is run with seeds 4101 and 4102, and the two outputs are averaged before regression. The sampled inputs are then converted into 12 features: total arrival rate, Class 1 arrival share, average levels, and Class 1 minus Class 2 gaps for balking step, balking threshold, no-show step, no-show threshold, and cancellation probability. Four targets are modeled separately: `average_utilization`, `overall_percent_served`, `mean_offered_booking_delay`, and `access_advantage_class_1`.

The regression is ordinary least squares on standardized variables. Each feature and each target is centered and scaled before fitting, so a coefficient is read as the change in target standard deviations associated with a one standard deviation increase in the feature, holding the other sampled features fixed. The sign gives direction; the absolute value ranks significant coefficients within each target.

Significance is assessed with `statsmodels` ordinary least squares and HC3 robust standard errors. The tested null hypothesis is that the partial standardized coefficient equals zero after controlling for the other 11 sampled features in this randomized design. A coefficient is treated as significant when

its robust p-value is below 0.05. This is a screening test for stable linear association in the simulation design, not proof of a causal effect in real appointment data.

Full-sample and 80/20 train-test R-squared values are saved in `outputs/reports/metric_analysis/data/regression_model_scores.csv` as a rough check that the linear screen is explaining meaningful variation.

Target metric	Significant feature	Std. coef.	95% HC3 CI	HC3 p
Utilization	Average no-show threshold	0.512	[0.433, 0.591]	< 0.001
Utilization	Average no-show step	-0.423	[-0.501, -0.346]	< 0.001
Utilization	Average cancellation probability	0.320	[0.247, 0.393]	< 0.001
Utilization	Total arrival rate	-0.109	[-0.188, -0.031]	0.006
Utilization	Average balking threshold	-0.086	[-0.162, -0.009]	0.028
Overall served rate	Total arrival rate	-0.774	[-0.855, -0.694]	< 0.001
Overall served rate	Average no-show threshold	0.251	[0.193, 0.308]	< 0.001
Overall served rate	Average no-show step	-0.175	[-0.232, -0.119]	< 0.001
Overall served rate	Average cancellation probability	0.117	[0.058, 0.176]	< 0.001
Offered wait	Total arrival rate	0.576	[0.499, 0.653]	< 0.001
Offered wait	Average cancellation probability	-0.441	[-0.506, -0.376]	< 0.001
Offered wait	Average balking threshold	0.278	[0.202, 0.353]	< 0.001
Offered wait	Average balking step	-0.206	[-0.291, -0.122]	< 0.001
Class gap	Cancellation probability gap	-0.452	[-0.558, -0.347]	< 0.001
Class gap	Balking threshold gap	0.429	[0.331, 0.528]	< 0.001
Class gap	Balking step gap	-0.349	[-0.431, -0.266]	< 0.001
Class gap	No-show threshold gap	0.268	[0.179, 0.357]	< 0.001
Class gap	No-show step gap	-0.211	[-0.286, -0.136]	< 0.001
Class gap	Class 1 arrival share	-0.120	[-0.209, -0.030]	0.009

Table 3: Significant regression drivers. Gap means Class 1 value minus Class 2 value.

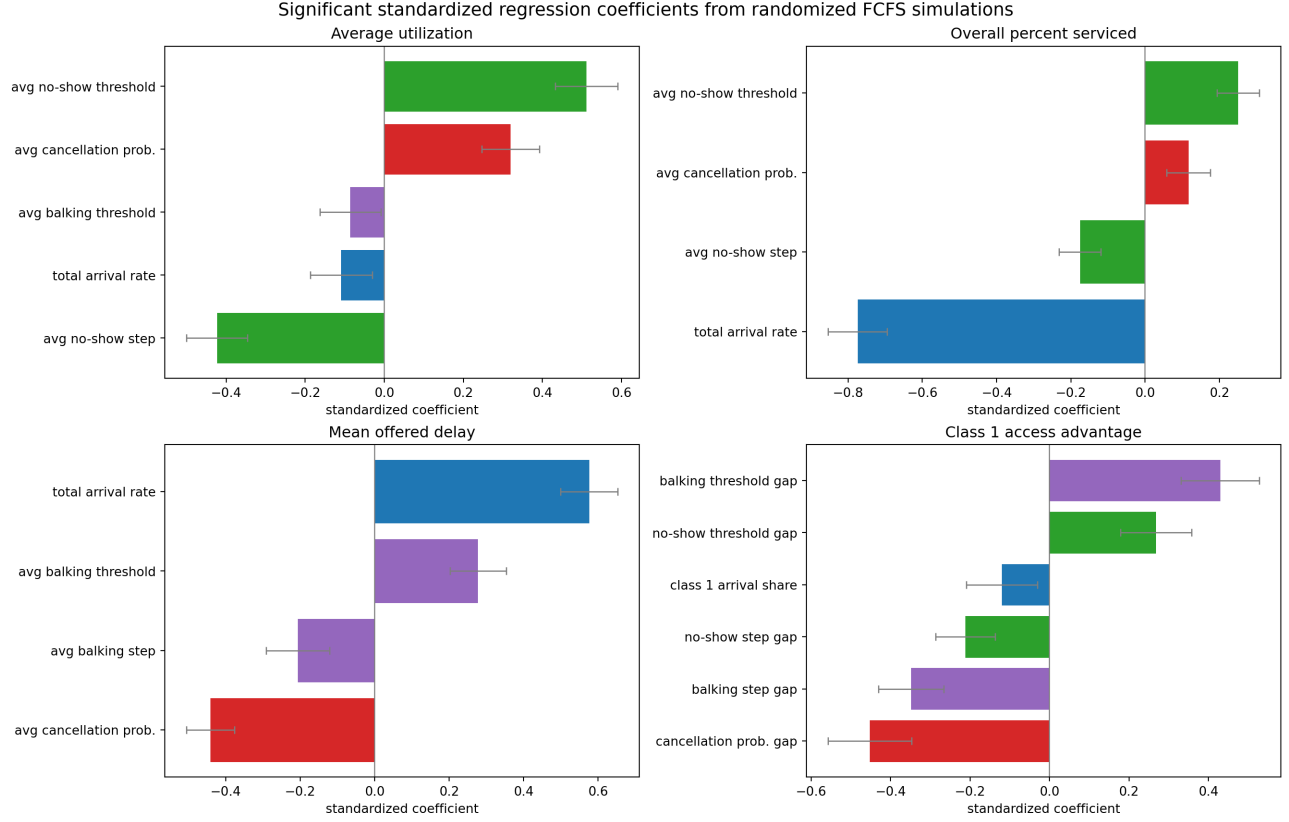


Figure 29: Significant standardized regression coefficient plot from randomized FCFS simulations. Error bars show 95% HC3 robust confidence intervals.

The regression screen supports the metric-by-metric reading: average no-show behavior explains utilization, total arrival rate explains access and wait, and class parameter gaps explain class advantage.

7 Final Takeaways

1. Use `overall_percent_served` as the main access metric and `average_utilization` as the capacity-use metric.
2. Do not interpret high utilization as good performance by itself. In the baseline, utilization is about 0.839 while only about 0.269 of arrivals complete service.
3. Use `mean_offered_booking_delay` for patient-facing wait because it includes patients who balk after seeing a long-delay offer.
4. Use balking rate as a diagnostic, not as a final success metric.
5. For class benefit, look at behavior gaps. Cancellation differences are the strongest class-gap driver in this run, with balking and no-show differences also important.

A Appendix: Exact Metric Definitions

Let i be patient class, D be measured days, S be slots per day, A_i be arrivals, B_i be booked patients, K_i be balked patients, C_i be canceled patients, H_i be no-shows, V_i be served patients, and $O_i = B_i + K_i$

be offered patients. Let $\bar{\tau}_i^{\text{offered}}$ be class i mean offered booking delay.

$$\begin{aligned}
\text{percent_serviced}_i &= \frac{V_i}{A_i}, \\
\text{overall_percent_serviced} &= \frac{\sum_i V_i}{\sum_i A_i}, \\
\text{mean_offered_booking_delay} &= \frac{\sum_i \text{total offered delay}_i}{\sum_i O_i}, \\
\text{mean_accepted_booking_delay} &= \frac{\sum_i \text{total accepted delay}_i}{\sum_i B_i}, \\
\text{average_utilization} &= \frac{1}{D} \sum_d \frac{\text{served slots}_d}{S}, \\
\text{balking_rate} &= \frac{\sum_i K_i}{\sum_i O_i}, \\
\Delta_{\text{access}} &= \text{percent_serviced}_1 - \text{percent_serviced}_2, \\
\Delta_{\text{delay}} &= \bar{\tau}_2^{\text{offered}} - \bar{\tau}_1^{\text{offered}}, \\
\Delta_{\text{balking}} &= \frac{K_1}{O_1} - \frac{K_2}{O_2}.
\end{aligned}$$

Positive Δ_{access} means Class 1 has the higher served rate. Positive Δ_{delay} means Class 1 has the lower offered wait. Positive Δ_{balking} means Class 1 rejects offers more often.

B Appendix: Dense Reference Plots

These figures are retained for traceability when the full diagnostic view is useful.

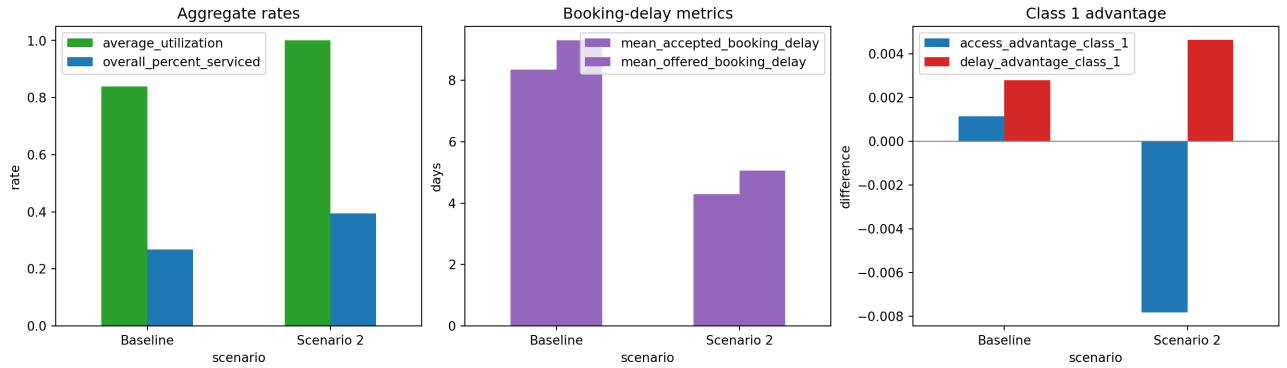


Figure 30: Scenario-level metric comparison.

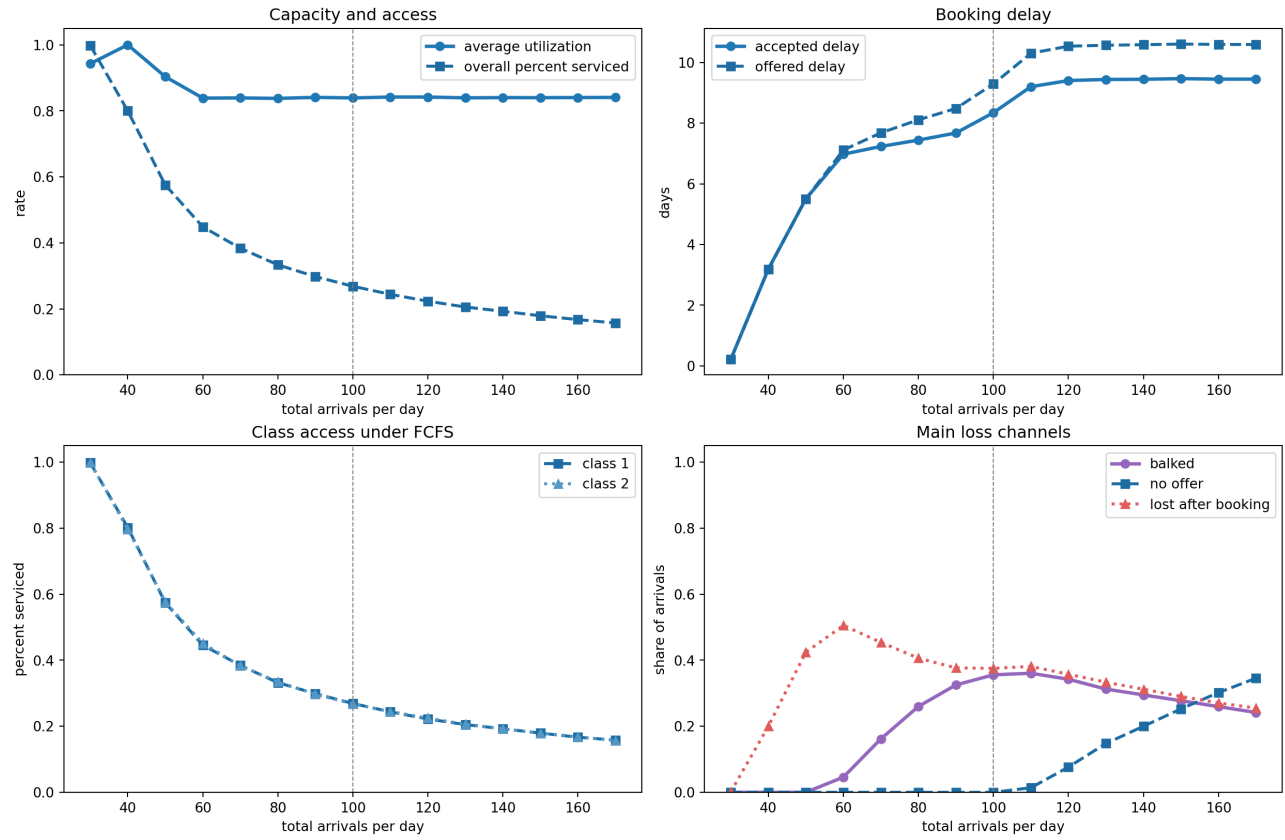


Figure 31: Full FCFS stress diagnostic plot.

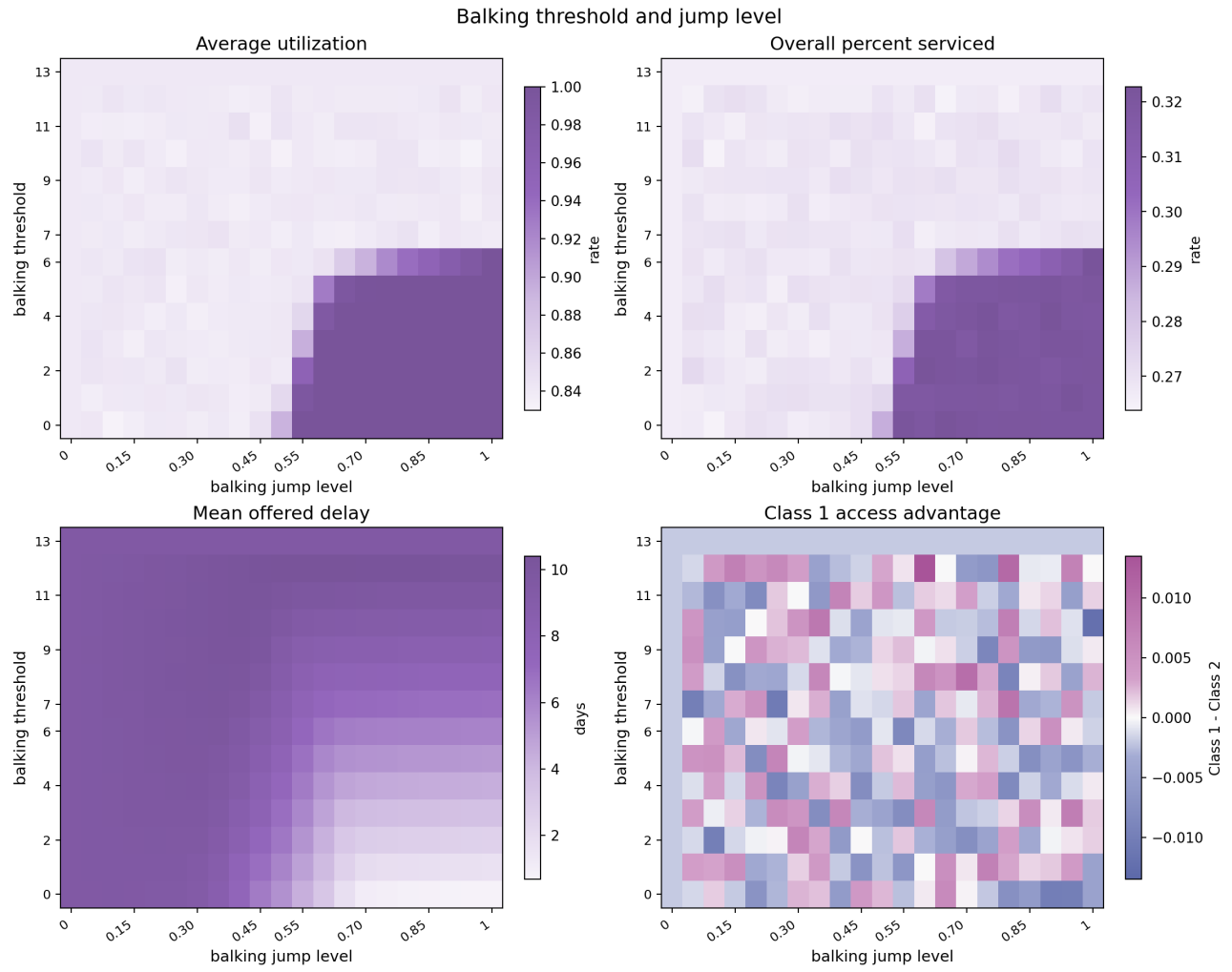


Figure 32: Balking threshold and jump level when both classes share the same values.

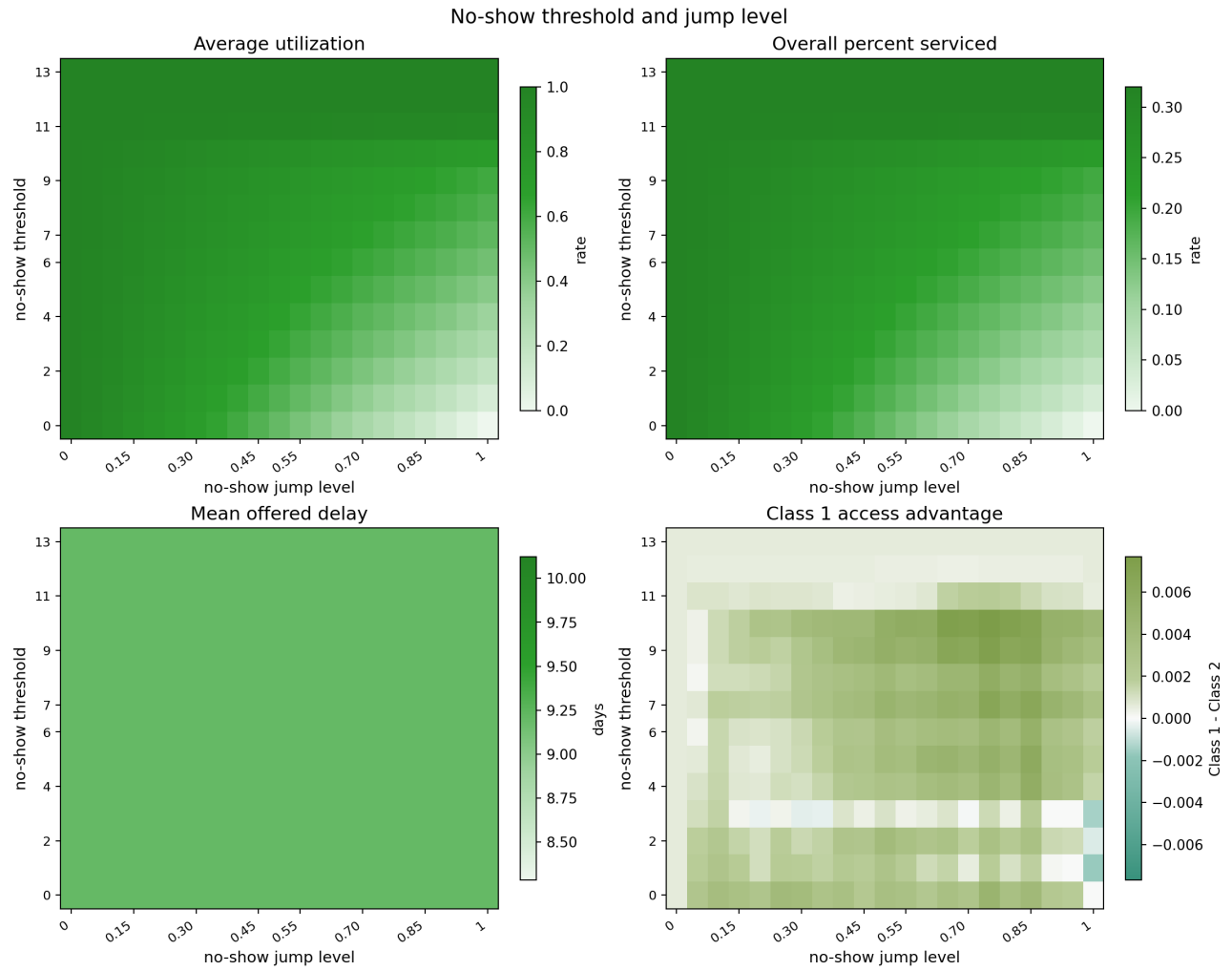


Figure 33: No-show threshold and jump level when both classes share the same values.